

RNA-SeqTutorial

February 03, 2026

Galaxy.msi.umn.edu

The screenshot displays the Galaxy/UMN web interface. The top navigation bar includes links for Analyze Data, Workflow, Shared Data, Visualization, Admin, Help, and User, along with a status indicator 'Using 90.2 Gb'. The left sidebar lists various tools and workflows, including 'Multiple Alignments', 'Metagenomic analyses', 'Metagenomics Mothur', 'FASTA manipulation', 'NCBI BLAST+', 'NGS: QC and manipulation', 'NGS: Picard (beta)', 'NGS: Assembly', 'NGS: Mapping', 'NGS: Indel Analysis', 'NGS: RNA Analysis', 'NGS: SAM Tools', 'NGS: GATK Tools', 'NGS: Peak Calling', 'NGS: Simulation', 'SNP/WGA: Data; Filters', 'SNP/WGA: QC; LD; Plots', 'SNP/WGA: Statistical Models', 'Human Genome Variation', 'VCF Tools', 'IGVTools', 'MSI', 'Masonic Cancer Center Tools', 'EMBOSS', and 'Workflows'. The main panel shows the 'Tophat for Illumina (version 1.5.0)' tool configuration. It includes fields for 'RNA-Seq FASTQ file:', 'Will you select a reference genome from your history or use a built-in index?:' (set to 'Use a built-in index'), 'Select a reference genome:' (set to 'Amellifera_Honeybee apiMel3'), 'Is this library mate-paired?:' (set to 'Single-end'), and 'TopHat settings to use:' (set to 'Use Defaults'). An 'Execute' button is at the bottom. A 'Tophat Overview' section provides a brief description of the tool. The right sidebar shows a 'History' panel with a list of previous jobs, including '14: Neighbor Joining Tree on data 12', '13: Neighbor Joining Tree on data 12', '12: hyphy.fasta', '11: Neighbor Joining Tree on data 6', '10: Neighbor Joining Tree on data 6', '9: pSymBGenesConcatenated.fasta', '8: Neighbor Joining Tree on data 6', '7: Neighbor Joining Tree on data 6', and '6:'. Each job entry has icons for viewing, deleting, and refreshing.

Galaxy / UMN

Analyze Data Workflow Shared Data Visualization Admin Help User Using 90.2 Gb

Tools Options ▾

Multiple Alignments

Metagenomic analyses

Metagenomics Mothur

FASTA manipulation

NCBI BLAST+

NGS: QC and manipulation

NGS: Picard (beta)

NGS: Assembly

NGS: Mapping

NGS: Indel Analysis

NGS: RNA Analysis

NGS: SAM Tools

NGS: GATK Tools

NGS: Peak Calling

NGS: Simulation

SNP/WGA: Data; Filters

SNP/WGA: QC; LD; Plots

SNP/WGA: Statistical Models

Human Genome Variation

VCF Tools

IGVTools

MSI

Masonic Cancer Center Tools

EMBOSS

Workflows

Tophat for Illumina (version 1.5.0)

RNA-Seq FASTQ file:

Nucleotide-space: Must have Sanger-scaled quality values with ASCII offset 33

Will you select a reference genome from your history or use a built-in index?:

Use a built-in index ▾

Built-ins were indexed using default options

Select a reference genome:

Amellifera_Honeybee apiMel3 ▾

If your genome of interest is not listed, contact the Galaxy team

Is this library mate-paired?:

Single-end ▾

TopHat settings to use:

Use Defaults ▾

You can use the default settings or set custom values for any of Tophat's parameters.

Execute

Tophat Overview

TopHat is a fast splice junction mapper for RNA-Seq reads. It aligns RNA-Seq reads to mammalian-sized genomes using the ultra high-throughput short read aligner Bowtie, and then analyzes the mapping results to identify splice junctions between exons. Please cite: Trapnell, C., Pachter, L. and Salzberg, S.L. TopHat: discovering splice junctions with RNA-

History Options ▾

imported: Unnamed 157.9 Mb history

14: Neighbor Joining Tree on data 12

13: Neighbor Joining Tree on data 12

12: hyphy.fasta

11: Neighbor Joining Tree on data 6

10: Neighbor Joining Tree on data 6

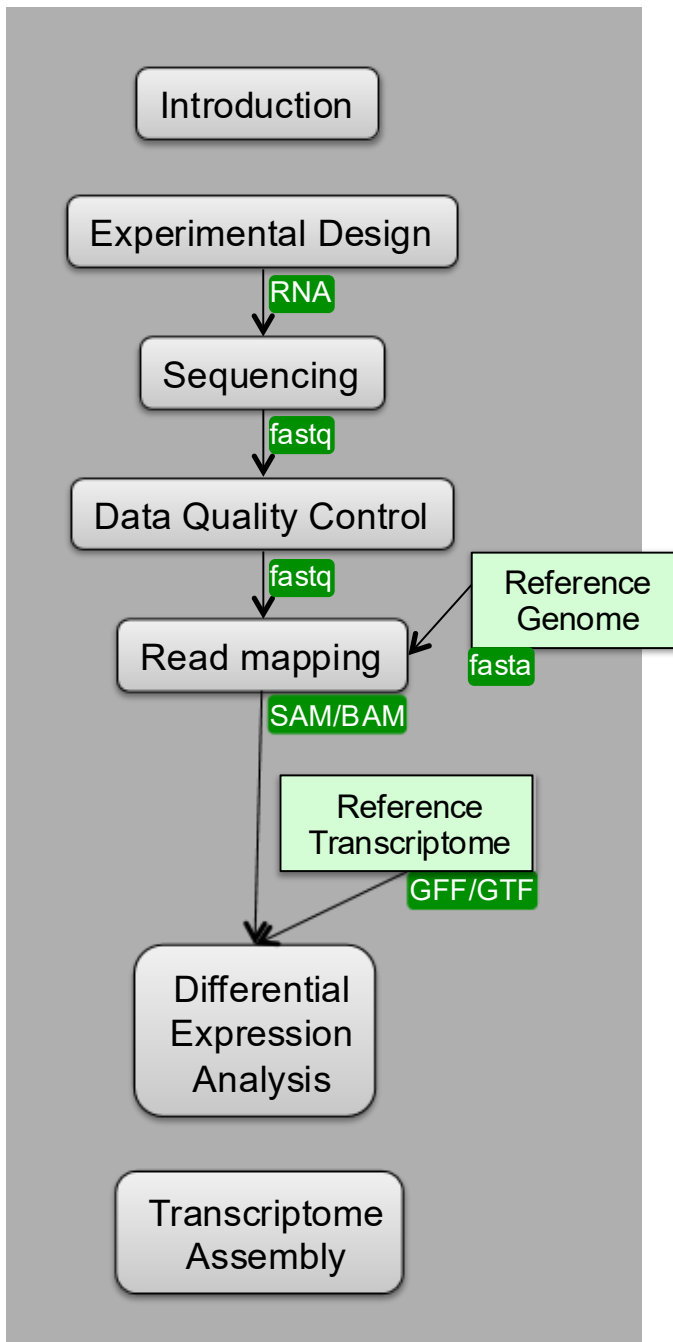
9: pSymBGenesConcatenated.fasta

8: Neighbor Joining Tree on data 6

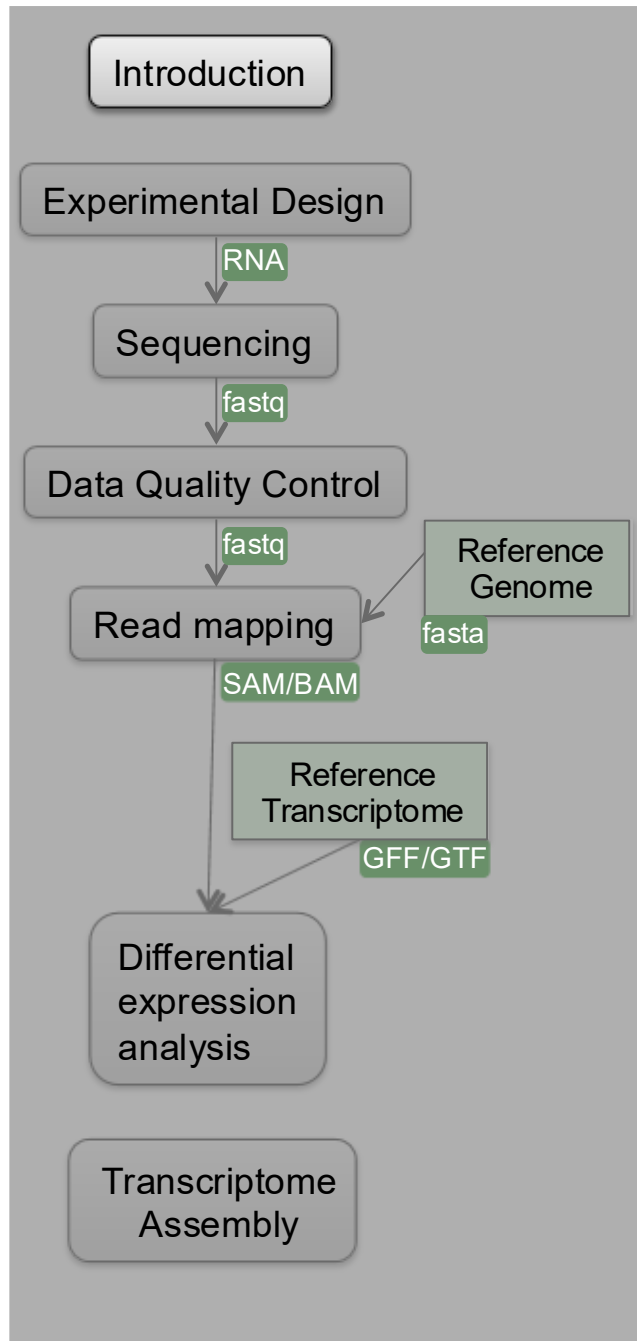
7: Neighbor Joining Tree on data 6

6:

Web-base platform for bioinformatic analysis



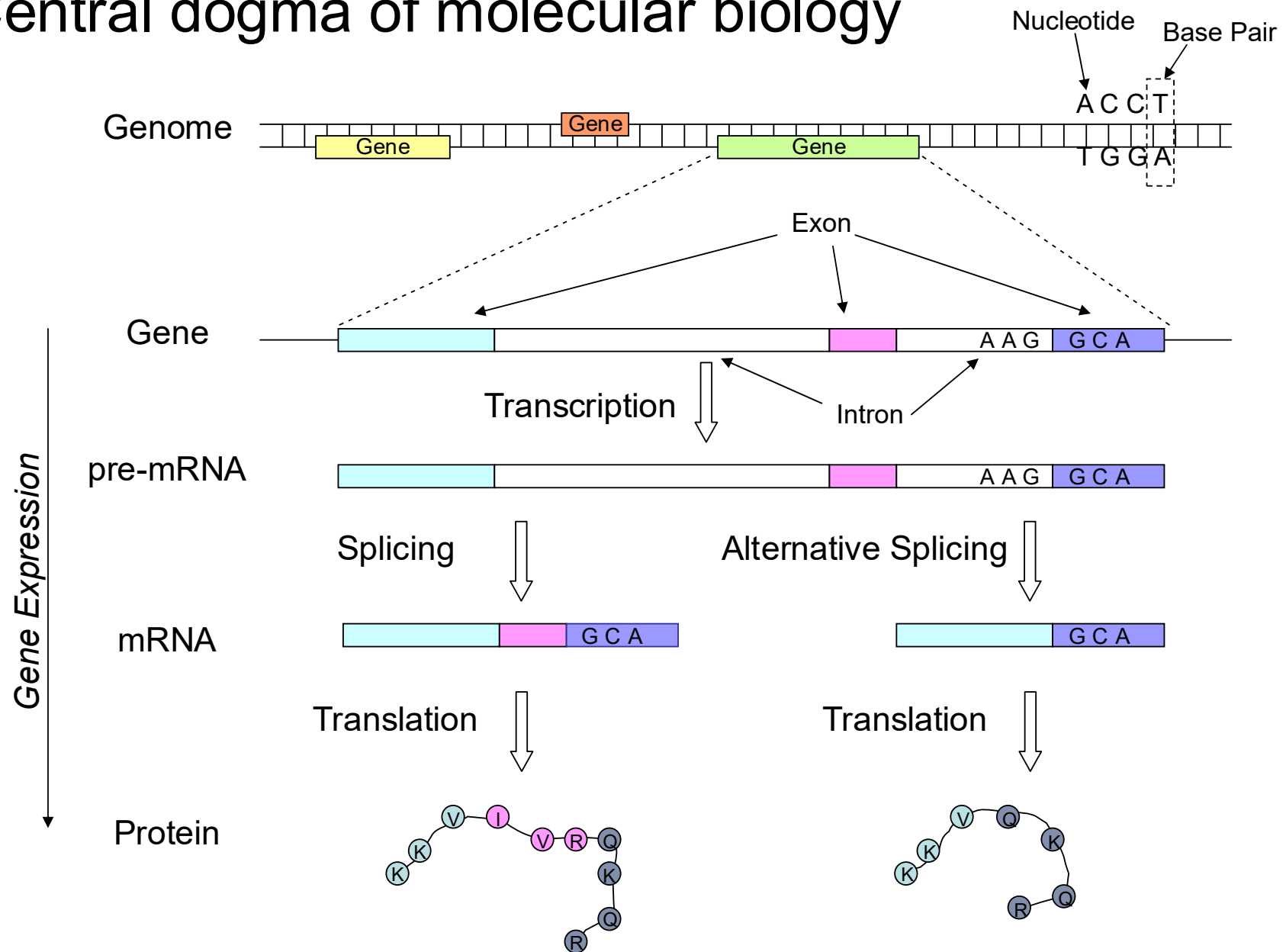
Outline



Introduction

- Geneexpression • RNA-Seq • Platform characteristics
- Microarray comparison

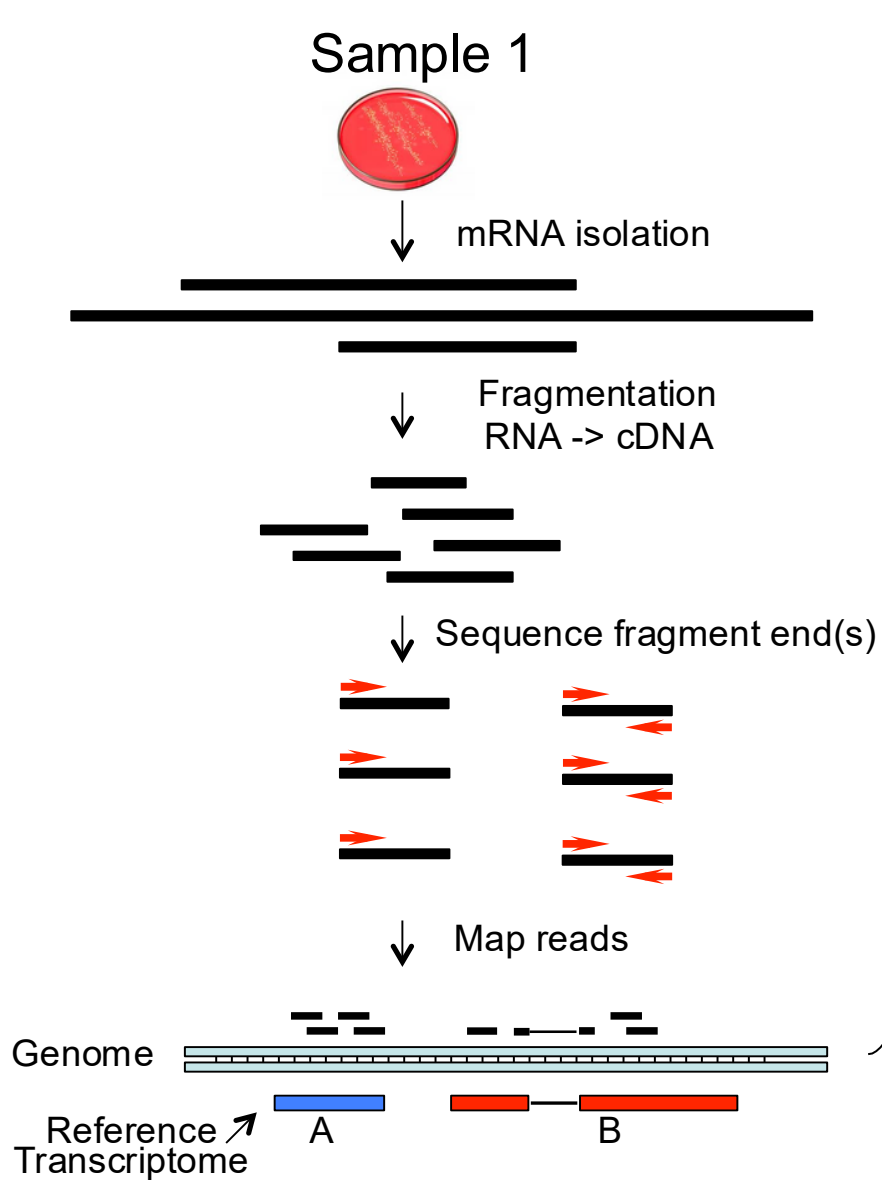
Central dogma of molecular biology



92–94% of human genes undergo alternative splicing,
86% with a minor isoform frequency of 15% or more
E. T. Wang, et al, Nature 456, 470-476 (2008)

Introduction

- RNA-Seq
 - High-throughput sequencing of RNA
 - Transcriptome assembly
 - Qualitative identification of expressed sequence
 - Differential expression analysis
 - Quantitative measurement of transcript expression



Calculate transcript abundance

	Gene A	Gene B
Sample 1	4	4

of Reads

	Gene A	Gene B
Sample 1	4	2

Reads per kilobase of exon

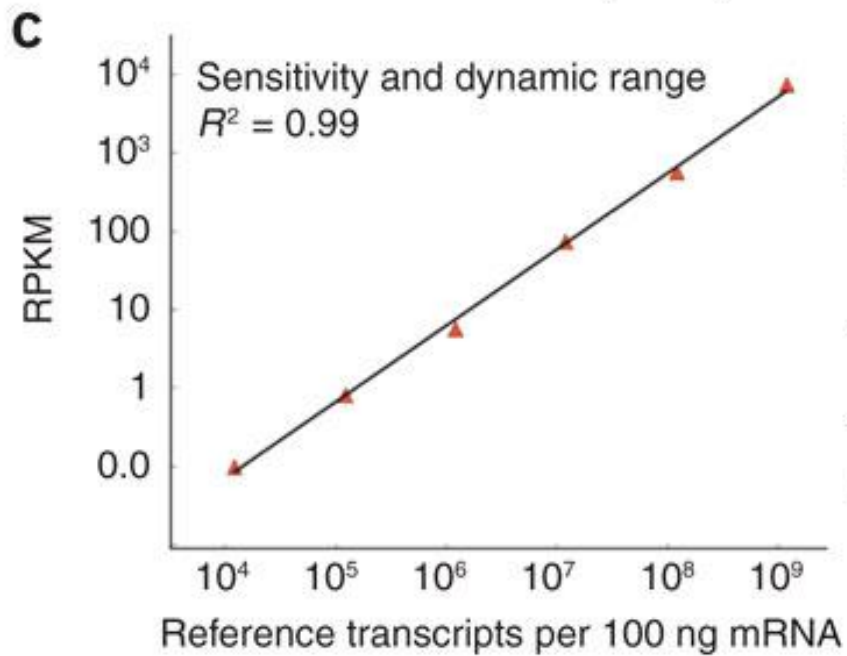
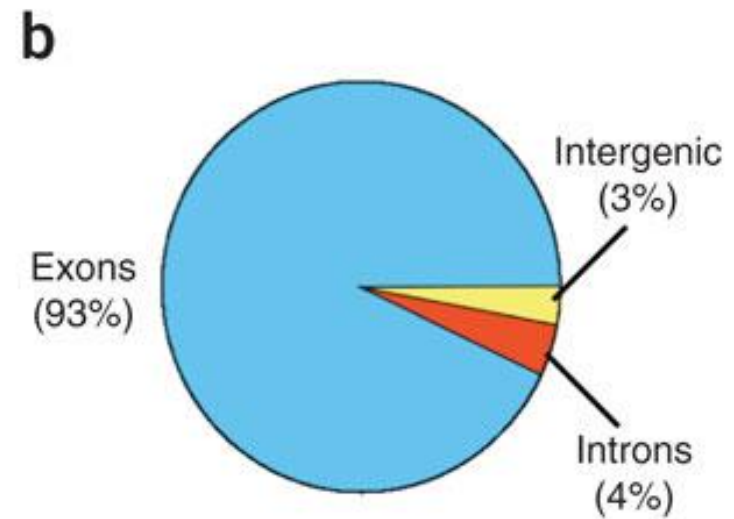
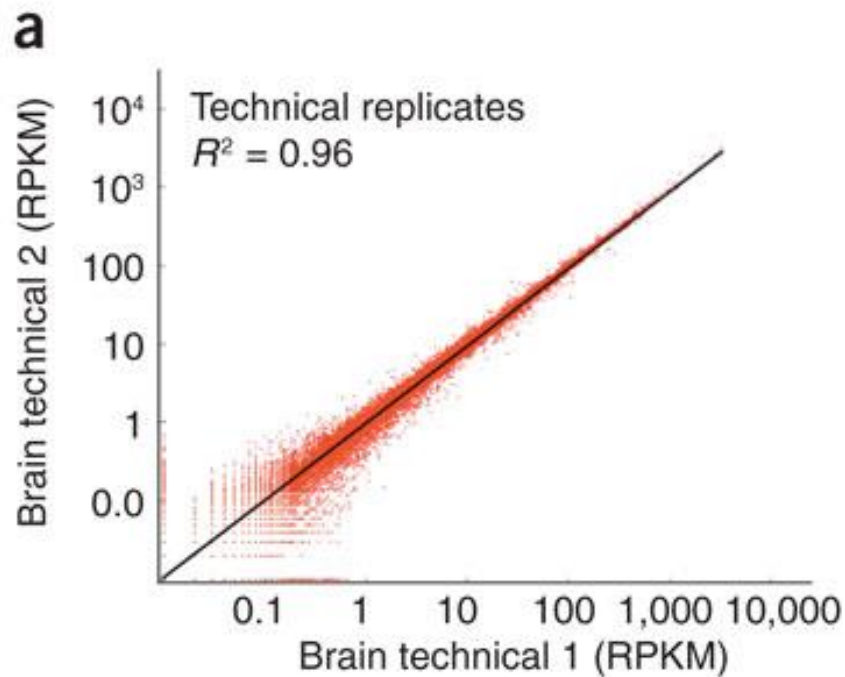
	Gene A	Gene B	Total
Sample 1	4 7	2 5	6 12
Sample 2			

Reads per kilobase of exon

	Gene A	Gene B	Total
Sample 1	.7	.3	6
Sample 2	.6	.3	12

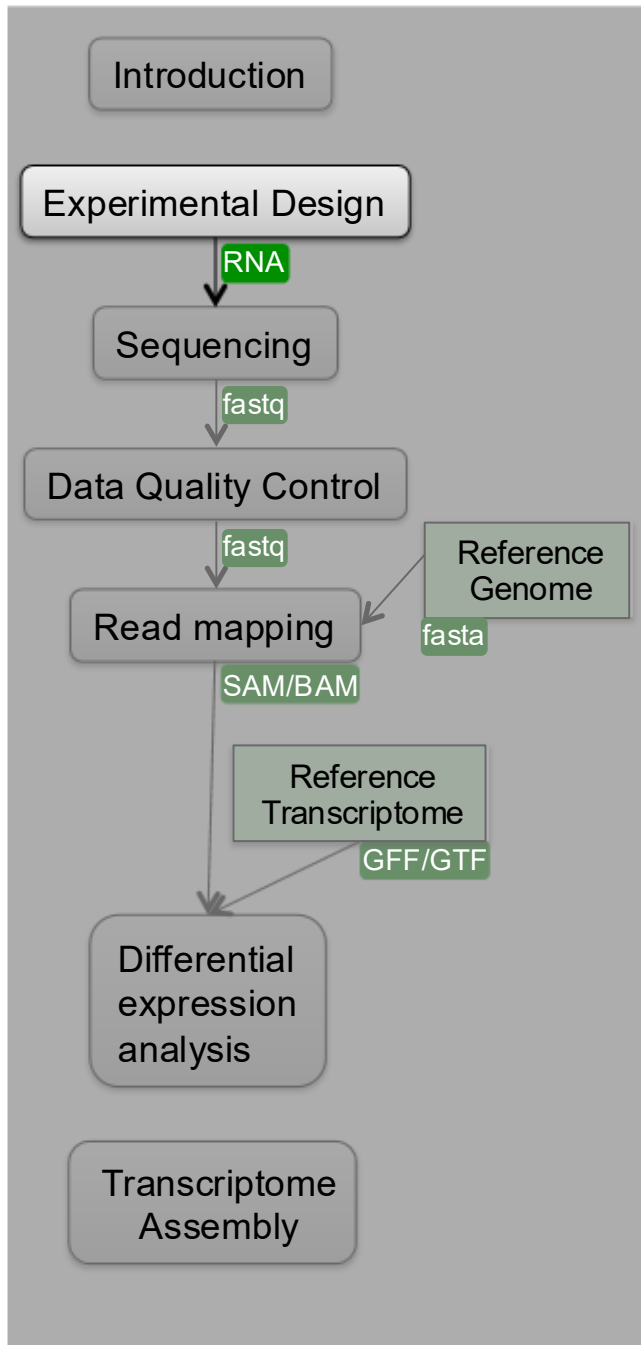
Reads per kilobase of exon per million mapped reads

RPKM



Introduction

- RNA-Seq (vs Microarray)
 - Strong concordance between platforms
 - Higher sensitivity and dynamic range –
 - Lower technical variation – Available for all species
 - Novel transcribed regions
 - Alternative splicing
 - Allele-specific expression
 - Fusion genes
 - Higher informatics cost

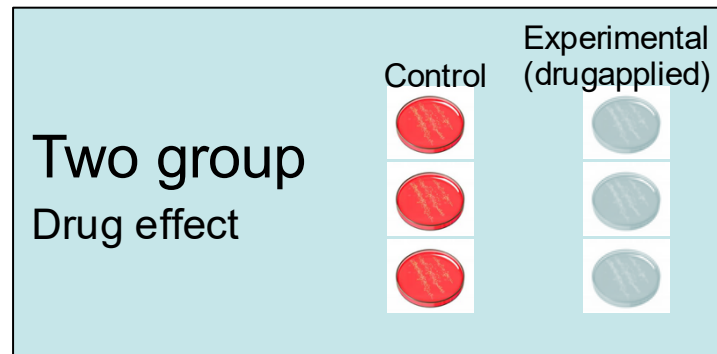


Experimental Design

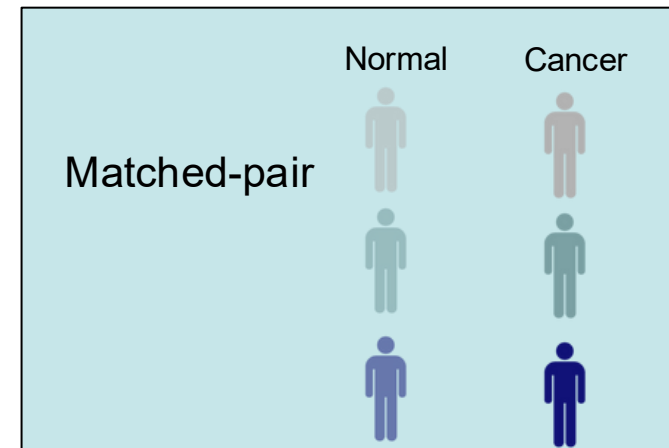
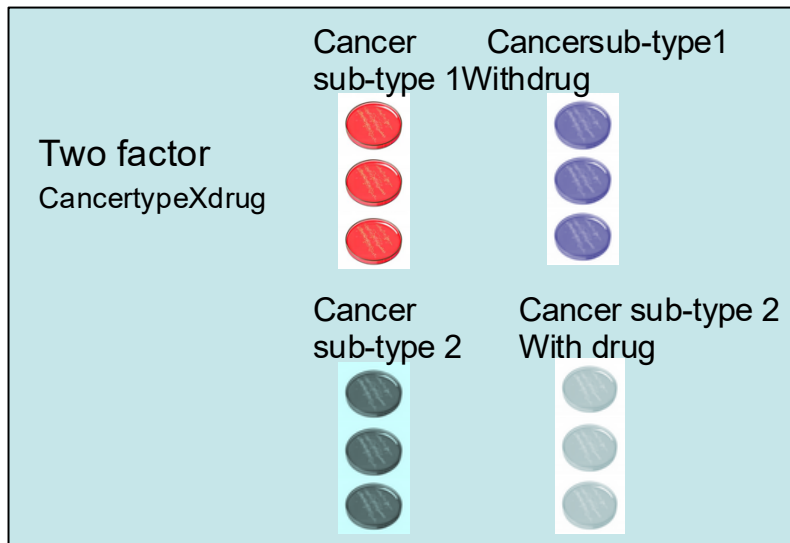
- Biological comparison(s)
- Paired-end vs single end reads
- Read length • Read depth
- Replicates • Pooling

Experimental design

- Simple designs (Pairwise comparisons)



- Complex designs –  Consult a statistician



Experimental design

- What are my goals?
 - Transcriptome assembly?
 - Differential expression analysis?
 - Identify rare transcripts?
- What are the characteristics of my system?
 - Large, complex genome? – Introns and high degree of alternative splicing? – No reference genome or transcriptome?

Experimental design

HiSeq 2000 Rates	Price Per Sample				
	10 million reads (1/20 lane)	20 million reads (1/10 lane)	50 million reads (1/4 lane)	100 million reads (1/2 lane)	200 million reads (1 lane)
Single-read (1x50 cycles)	\$267	\$345	\$581	\$975	\$1,762
Single-read (1x100 cycles)	\$290	\$395	\$696	\$1,205	\$2,225
Paired-end read (2x50 cycles)	\$320	\$432	\$835	\$1,480	\$2,775
Paired-end read (2x100 cycles)	\$365	\$540	\$1,050	\$1,940	\$3,700

BMGC RNA-Seq Price list

Experimental design

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10 million reads per sample, 50bp single-end reads

- Small genomes with no alternative splicing

Experimental design

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20 million reads per sample, 50bp paired-end reads

- Mammalian genomes (largetranscriptome,alternativesplicing,geneduplication)

Experimental design

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50-200 million reads per sample, 100bp paired-end reads

- TranscriptomeAssembly(100Xcoverageoftranscriptome)

50bp Paired-end >> 100bp Single-end

Experimental design

- Technical replicates
 - Not needed: low technical variation
 - Minimize batch effects
 - Randomize sample order 
- Biological replicates
 - Not needed for transcriptome assembly –
 - Essential for differential expression analysis –
 - Difficult to estimate
 - 3+ for cell lines • 5+ for inbred lines • 20+ for human samples

Experimental design

- Pooling samples
 - Limited RNA obtainable
 - Multiple pools per group required
 - Transcriptome assembly

Experimental design

RNA-seq: technical variability and sampling

Lauren M McIntyre, Kenneth K Lopiano, Alison M Morse, Victor Amin, Ann L Oberg, Linda J Young and Sergey V Nuzhdin
BMC Genomics 2011, 12:293

Statistical Design and Analysis of RNA Sequencing Data

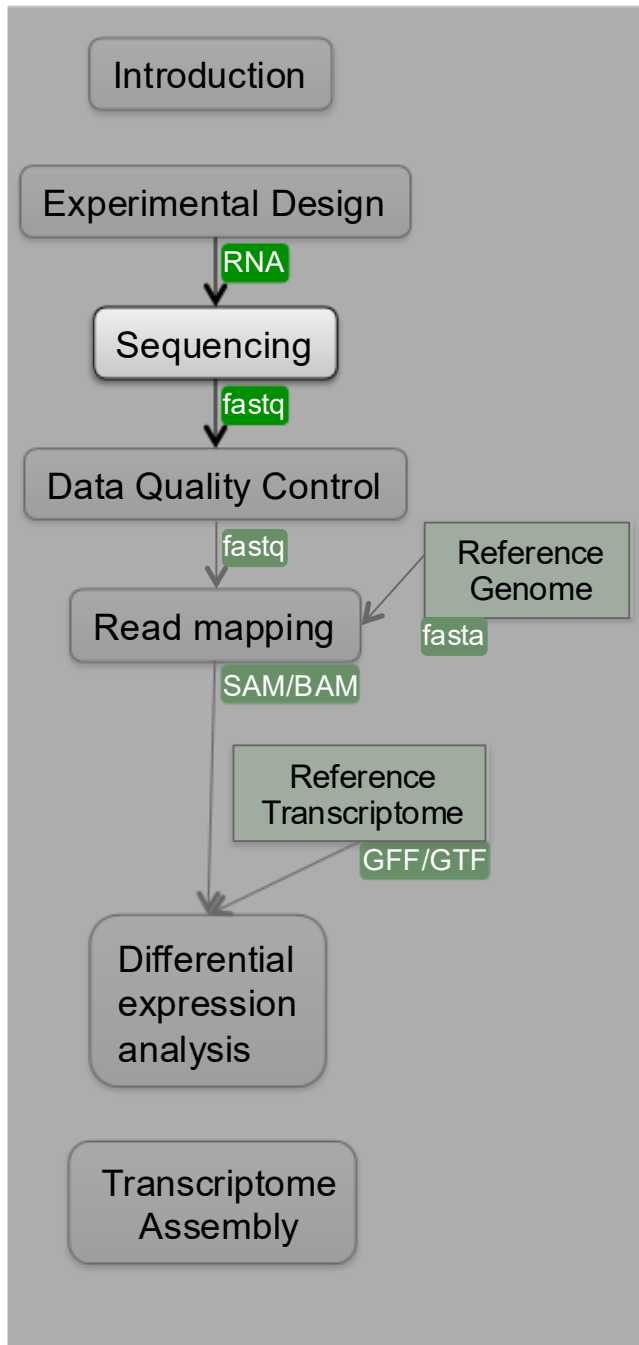
Paul L. Auer and R. W. Doerge Genetics. 2010 June; 185(2): 405–416.

Analyzing and minimizing PCR amplification bias in Illumina sequencing libraries

Daniel Aird, Michael G Ross, Wei-Sheng Chen, Maxwell Danielsson, Timothy Fennell, Carsten Russ, David B Jaffe, Chad Nusbaum and Andreas Gnirke
Genome Biology 2011, 12:R18

ENCODE RNA-Seq guidelines

http://www.encodeproject.org/ENCODE/experiment_guidelines.html



Sequencing

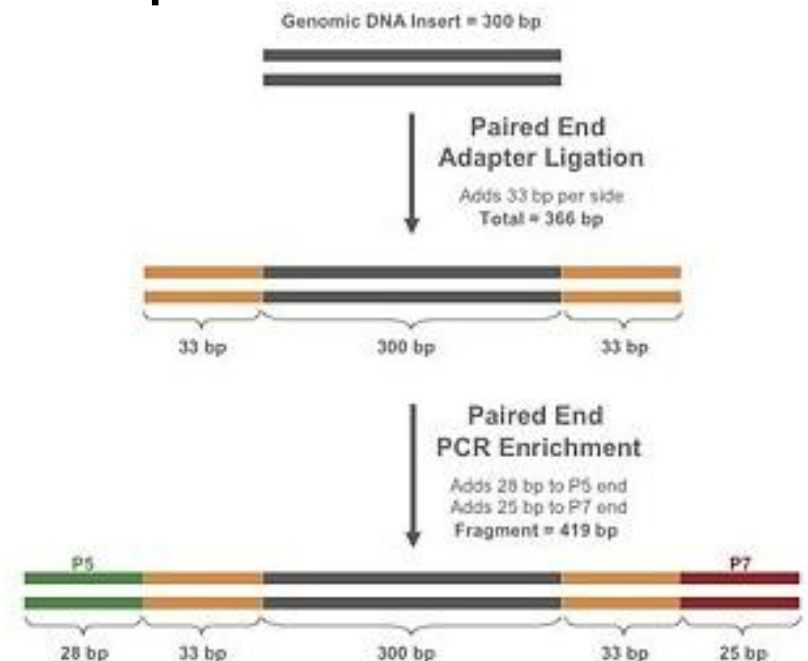
- Platforms • Library preparation
- Multiplexing
- Sequence reads

Sequencing

- Illumina sequencing by synthesis
 - GAIIx
 - replaced by HiSeq
 - HiSeq2000
 - MiSeq
 - low throughput, fast turnaround
- SOLiD (not available at BMGC)
 - “Color-space” reads (require special mapping software)
 - Low error rate
- 454 pyrosequencing
 - Longer reads, lower throughput

Sequencing

- Library preparation (IlluminaTruSeqprotocol for HiSeq)
 - RNA isolation
 - Poly-A purification
 - Fragmentation
 - cDNA synthesis using random primers
 - Adapter ligation
 - Size selection
 - PCR amplification

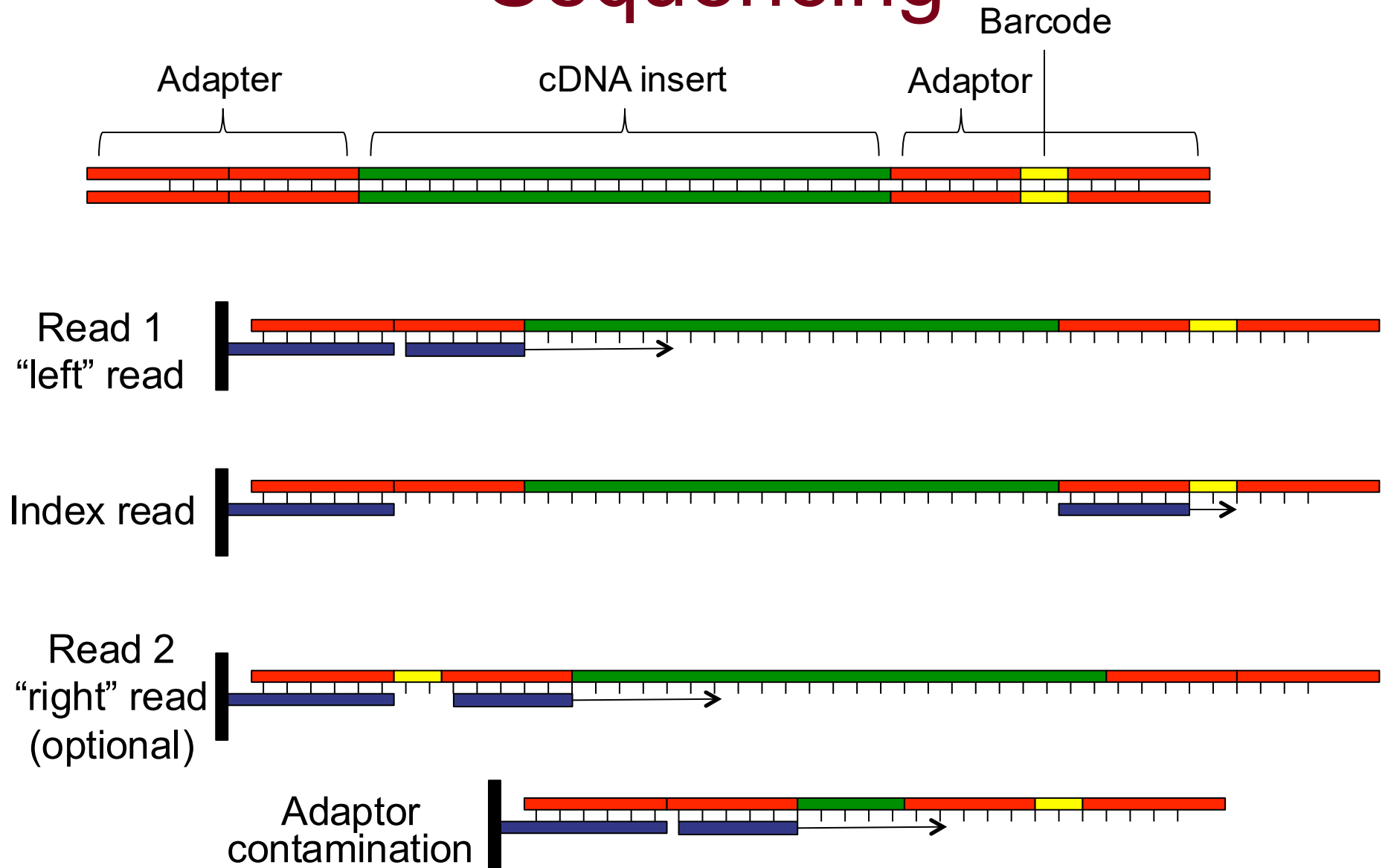


Sequencing

- Flowcell
 - 8lanes – 200 Million reads per lane – Multiplex up to 24 samples on one lane using barcodes



Sequencing



Sequencing

- Library types
 - Polyadenylated RNA > 200bp (standard method)
 - Total RNA
 - Small RNA
 - Strand-specific
 - Gene-dense genomes (bacteria, archaea, lower eukaryotes)
 - Antisense transcription (higher eukaryotes)
 - Low input –
Library capture

Sequence Data Format

- Datadelivery

- /project/PI-groupname/120318_SN261_0348_A81JUMABXX
 - fastq_fit/ Badreadsremoved by Illuminasoftware,for useindataanalysis
 - fastq/ Rawsequence output for submission to public archives, contains bad reads
- UploadtoGalaxy



- File names

- L1_R1_CCAAT_cancer1.fastq –
- L1_R2_CCAAT_cancer1.fastq

- Fastq format (Illumina Casava 1.8.0) – 4linesperread

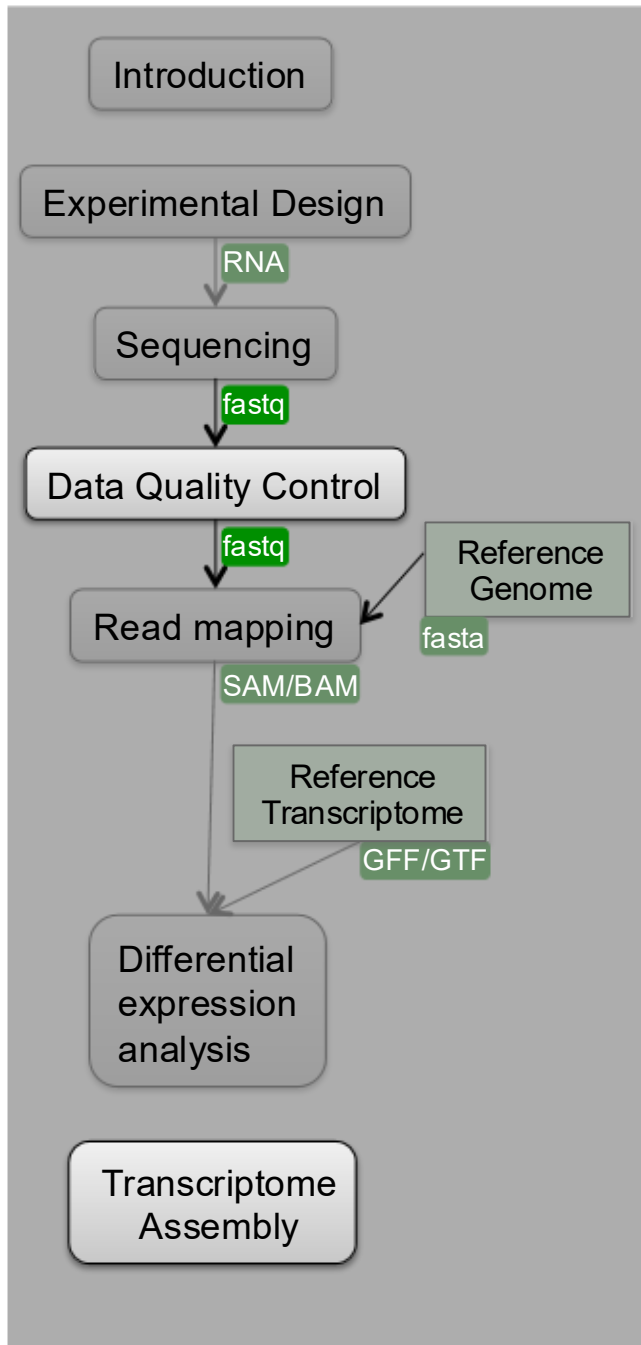


Machine ID
Read ID → @HWI-M00262:4:000000000-A0ABC:1:1:18376:2027 1:N:0:AGATC
Sequence + → TTCAGAGAGAATGAATTGTACGTGCTTTTTTTTGT
Quality score → +
Phred+33 → =1:?7A7+?77+<<@AC<3<,33@A;<A?A=:4=

QC Filter flag
Y=bad
N=good
barcode
Read pair #

Data Quality Control

- Quality assessment
- Trimming and filtering



Data Quality Assessment

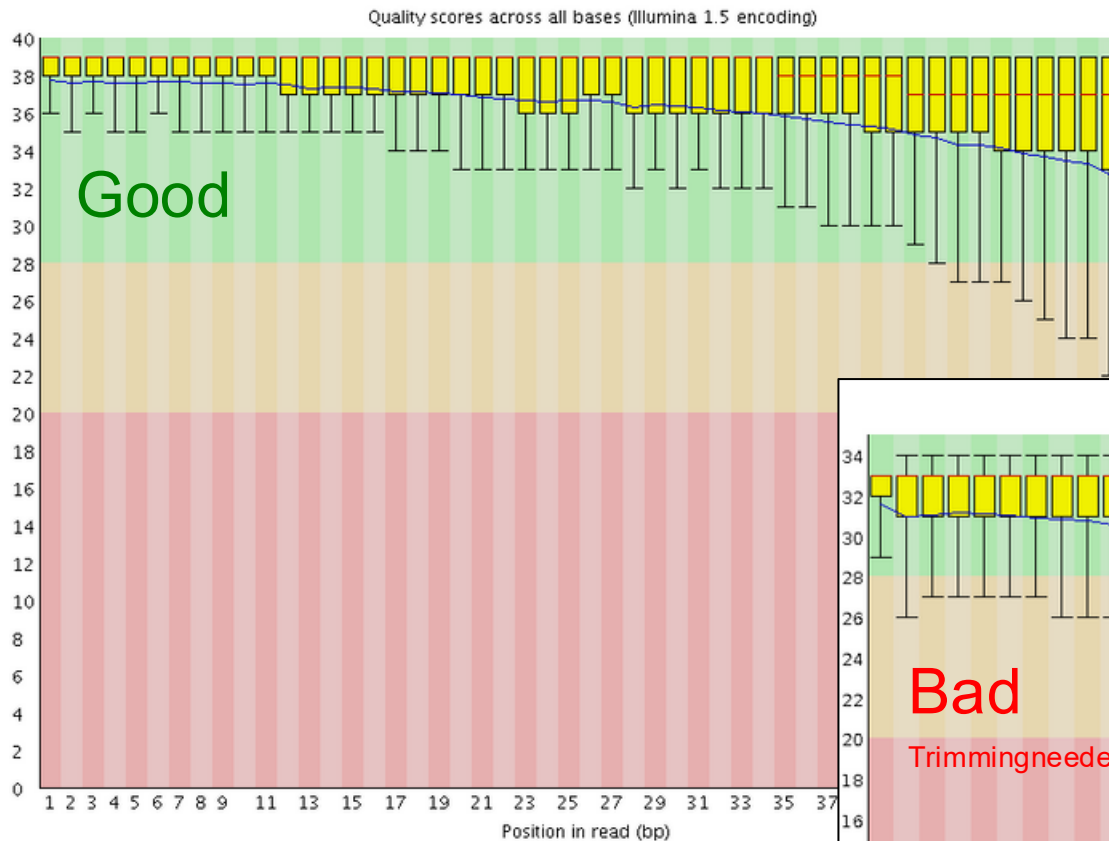
- Evaluate read library quality
 - Identify contaminants
 - Identify poor/bad samples
- Software
 - FastQC (recommended)
 - Command-line, Java GUI, or Galaxy
 - SolexaQC
 - Command-line • Supports quality-based read trimming and filtering
 - SAMStat
 - Command-line • Also works with bam alignment files

Data Quality Assessment

- Trimming: remove bad bases from (end of) read
 - Adaptor sequence
 - Low quality bases
- Filtering: remove bad reads from library
 - Low quality reads –
 - Contaminating sequence – Low
 - complexity reads (repeats) – Short
 - reads
 - Short (<20bp) reads slow down mapping software
 - Only needed if trimming was performed
- Software
 - Galaxy, many options (NGS: QC and manipulation) –
 - Tagdust – Many others:

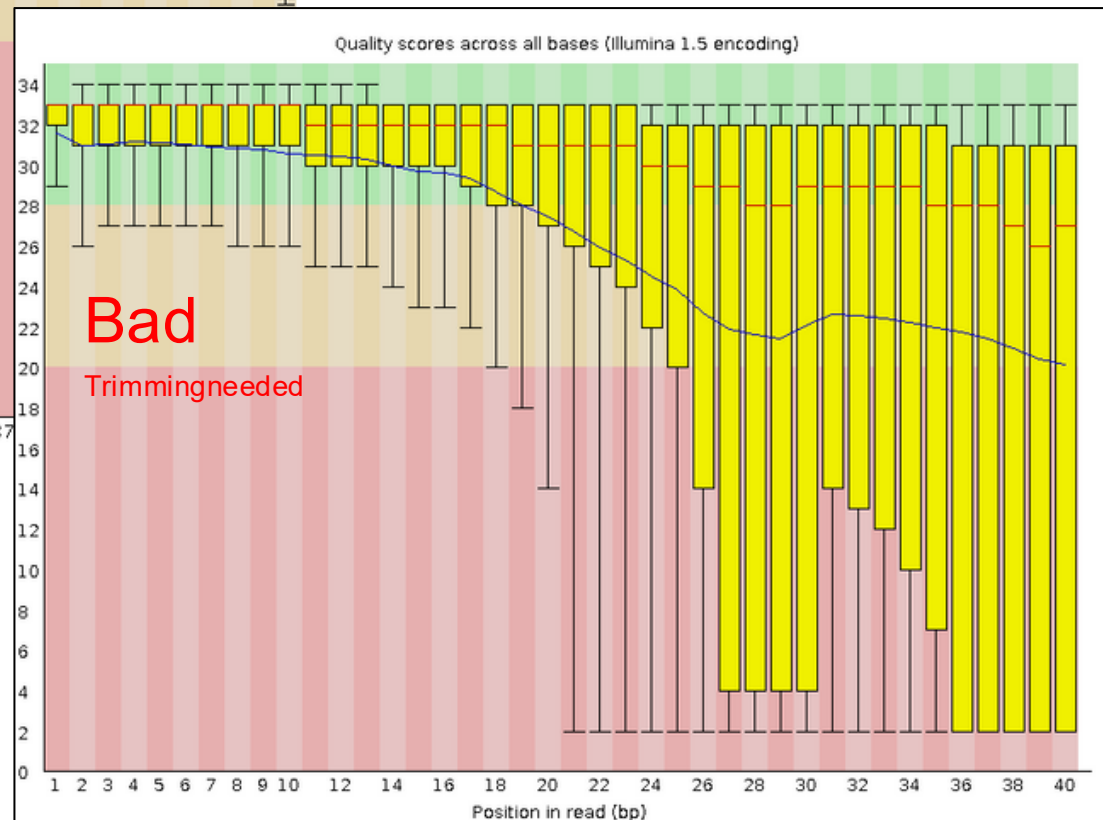
<http://seqanswers.com/wiki/Software/list>

Data Quality Assessment - FastQC

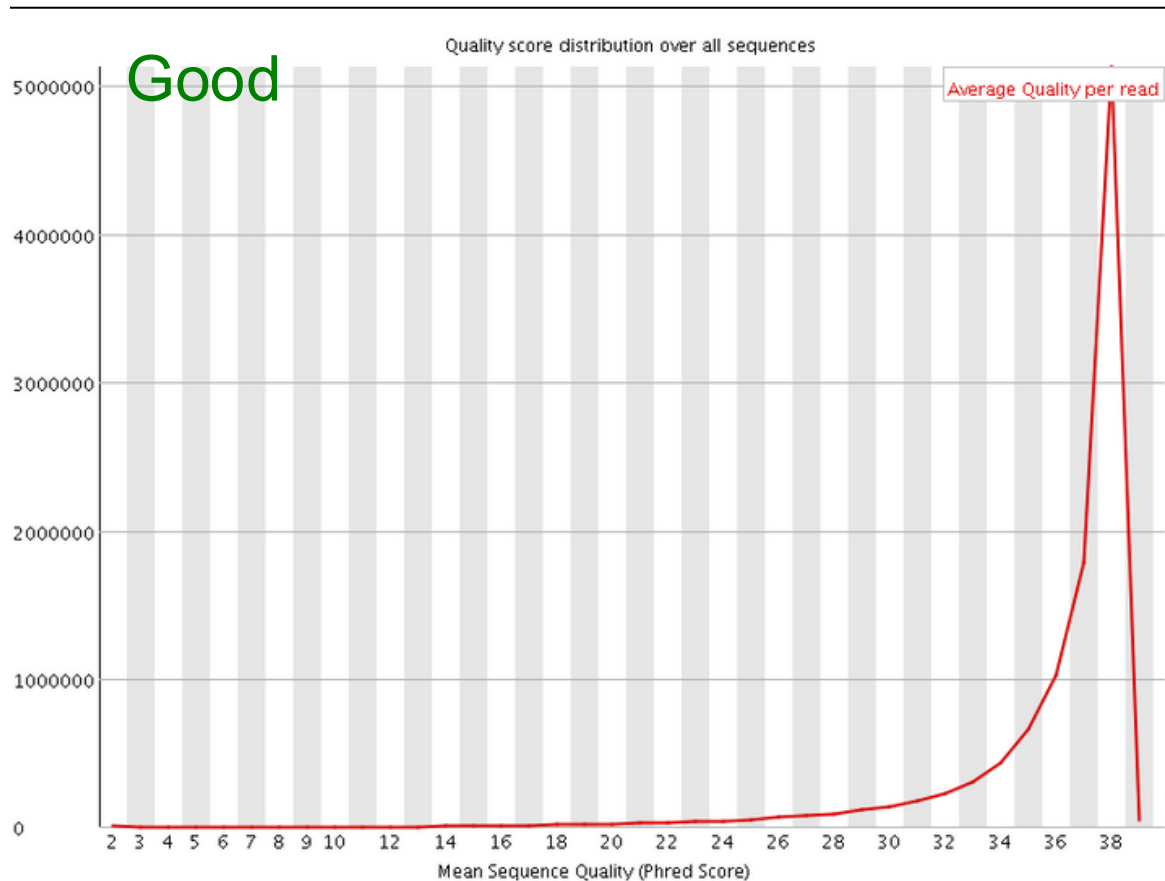


Quality scores across bases

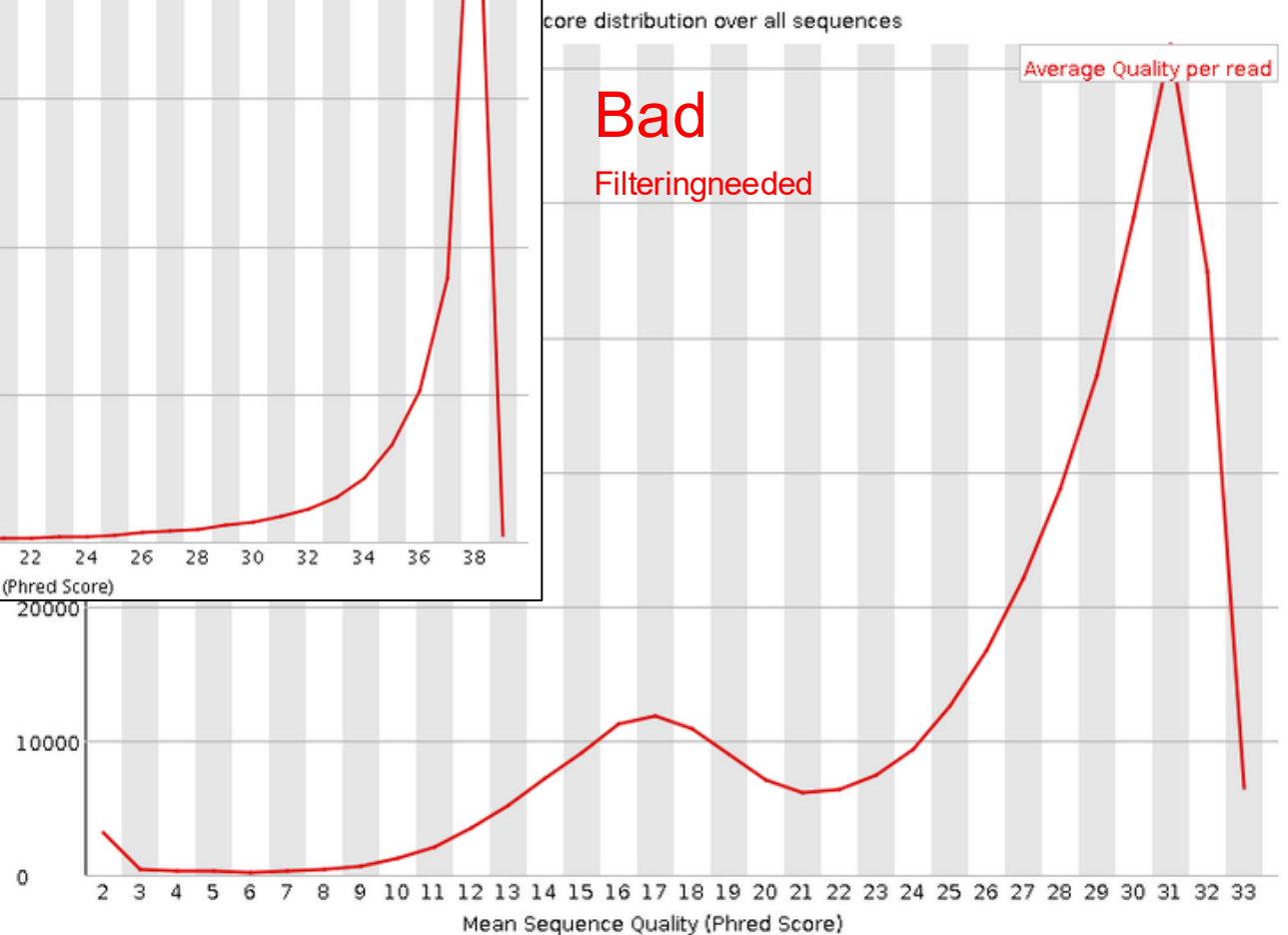
Phred 30 = 1 error / 1000 bases
Phred 20 = 1 error / 100 bases



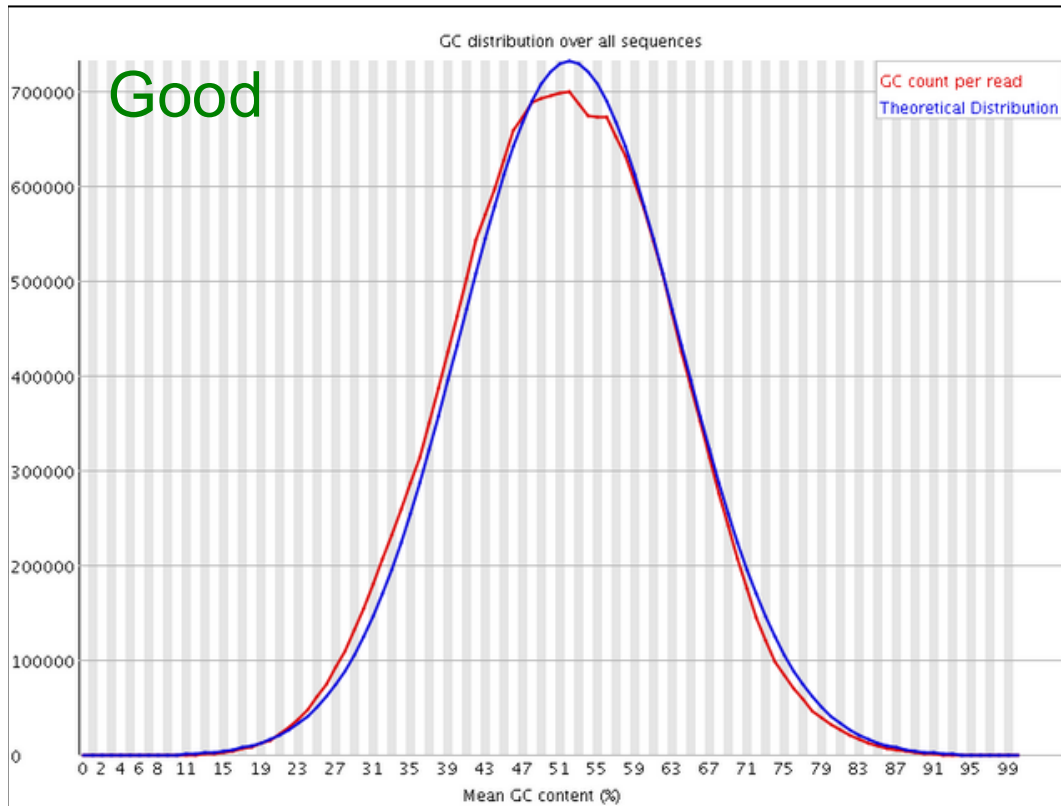
Data Quality Assessment - FastQC



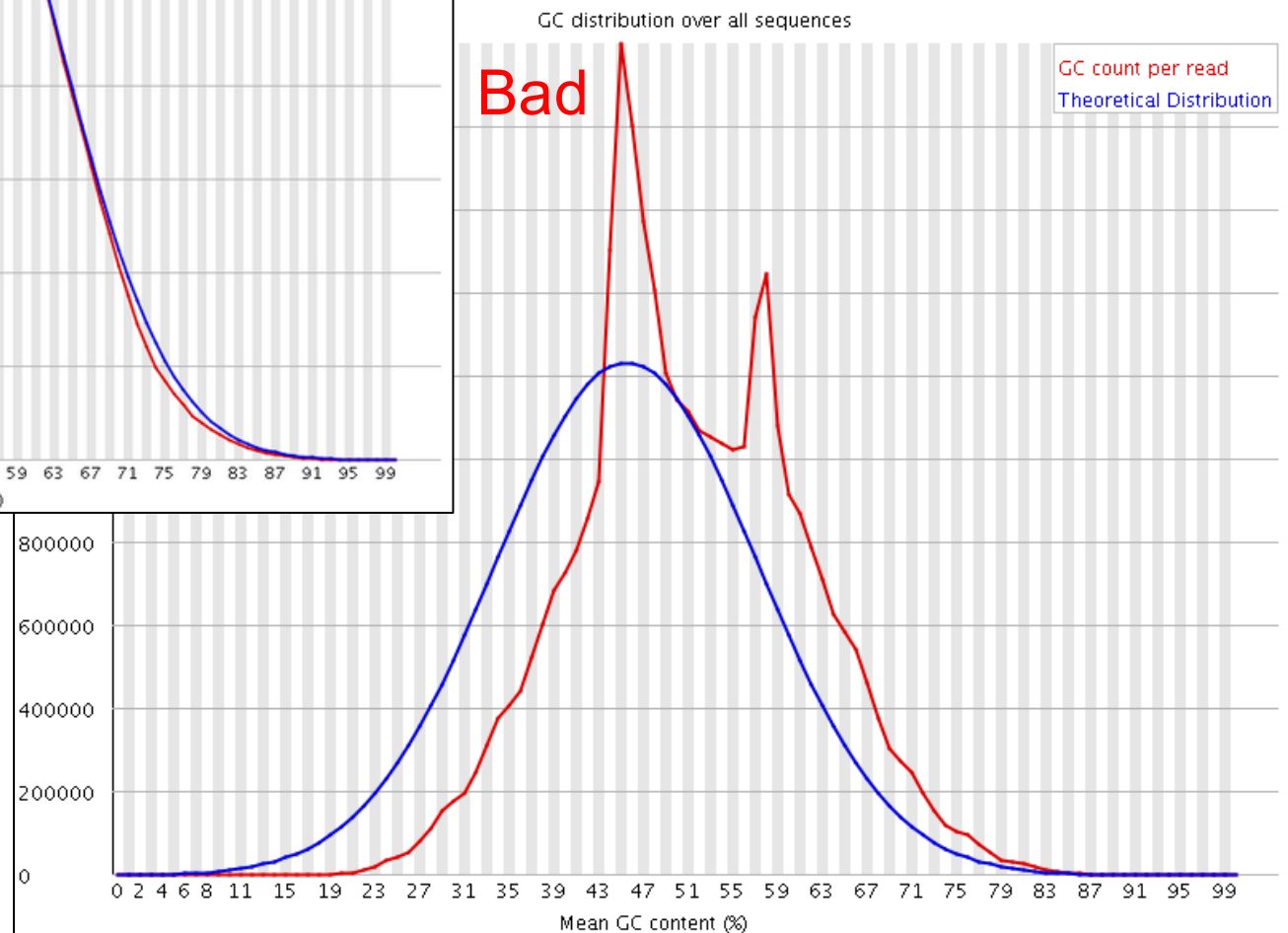
Quality scores across reads



Data Quality Assessment - FastQC



GC Distribution



Data Quality Assessment - FastQC

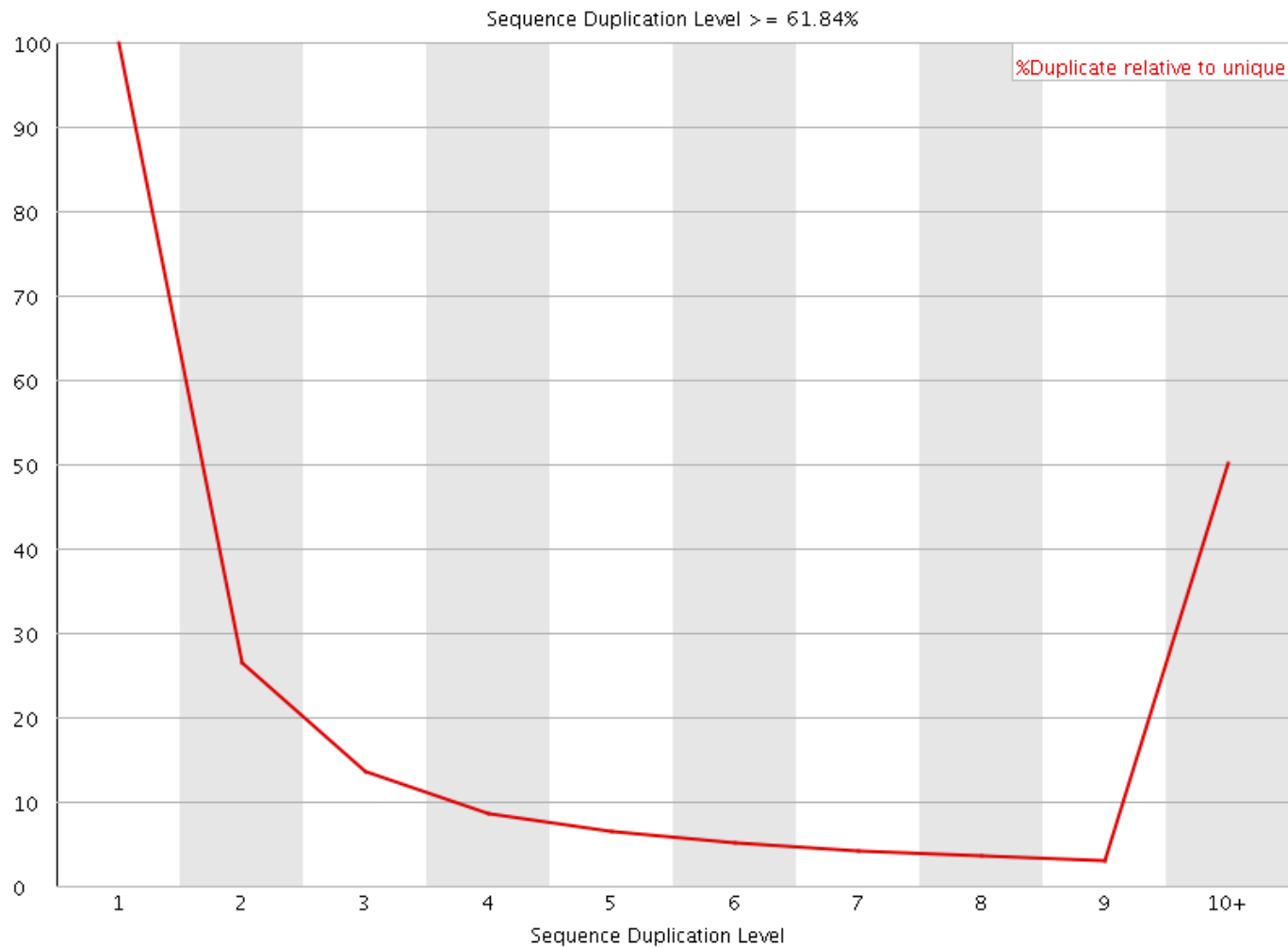
High level of sequencing adapter contamination, trimming needed

Overrepresented sequences

Sequence	Count	Percentage	Possible Source
GTATTACAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCG	820428	2.8366639370528275	Illumina Paired End PCR Primer 2 (100% over 43bp)
GTATACAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGT	749728	2.5922157461699773	Illumina Paired End PCR Primer 2 (100% over 44bp)
CGGTTCAGCAGGAATGCCGAGATCGGAAGAGCGGTTCAGCAGGAATGCCG	648852	2.243432780066747	Illumina Paired End Adapter 2 (100% over 31bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGATCGGAAGAGCGGTTCAG	176765	0.6111723403310748	Illumina Paired End PCR Primer 2 (97% over 36bp)
ACGTCGTAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCG	143840	0.4973327832615156	Illumina Paired End PCR Primer 2 (100% over 43bp)
GTATTACAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGT	124281	0.42970672717272257	Illumina Paired End PCR Primer 2 (100% over 44bp)
GTATCAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGTA	99207	0.34301232917842867	Illumina Paired End PCR Primer 2 (100% over 45bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGTATGCCGT	96289	0.33292322279941655	Illumina Paired End PCR Primer 2 (100% over 50bp)
CGGAAGAGCGGTTCAGCAGGAATGCCGAGATCGGAAGAGCGGTTCAGCAG	93842	0.3244626185124245	Illumina Paired End PCR Primer 2 (96% over 33bp)
CGTTACGAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCG	75370	0.26059491013918545	Illumina Paired End PCR Primer 2 (100% over 43bp)
CGTACGAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGT	63691	0.22021428183196043	Illumina Paired End PCR Primer 2 (100% over 44bp)
ACGTAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGTAT	56765	0.19626734873359242	Illumina Paired End PCR Primer 2 (100% over 46bp)
TACTGTAAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCG	42991	0.14864317078139472	Illumina Paired End PCR Primer 2 (100% over 43bp)

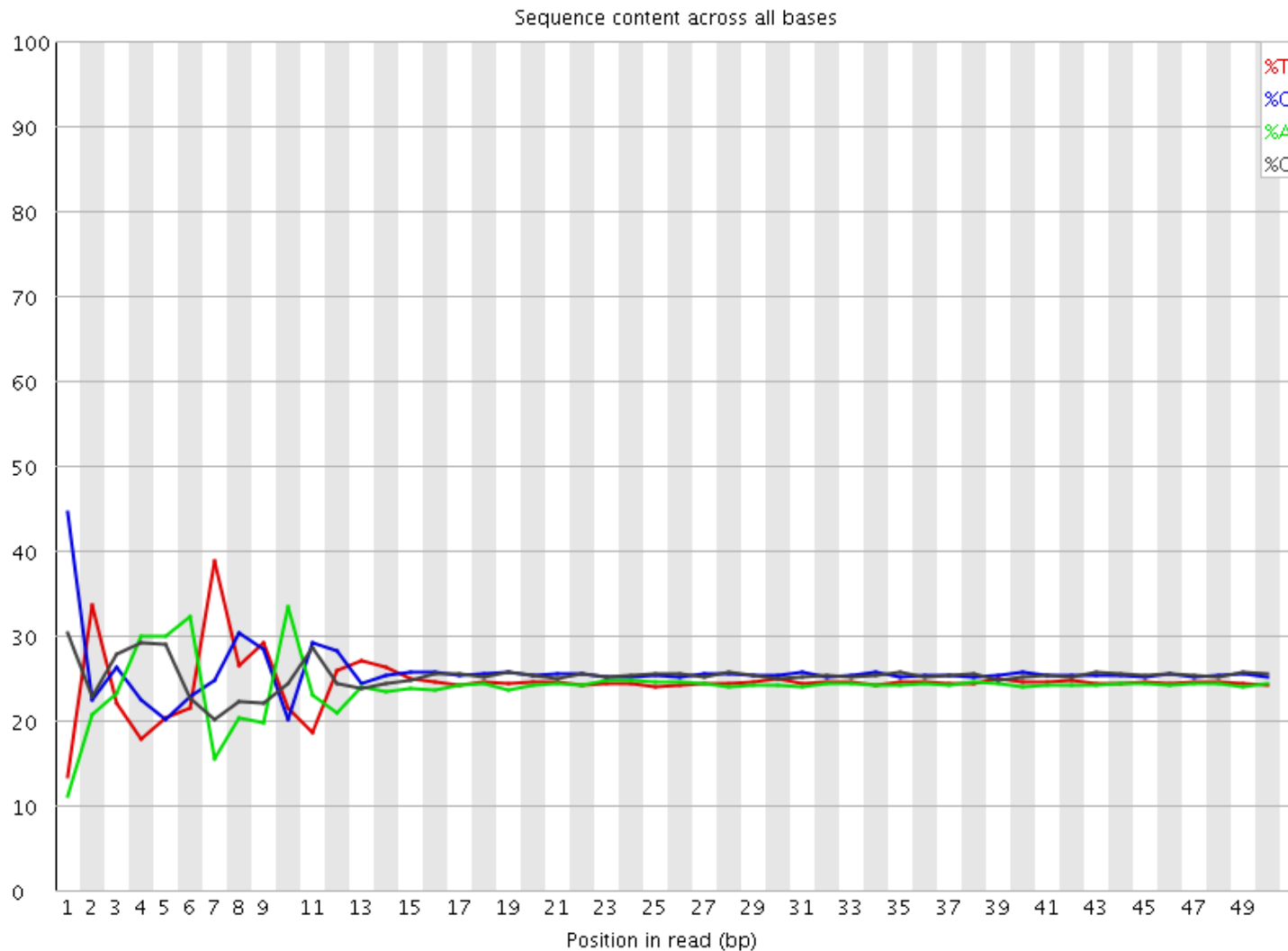
Data Quality Assessment - FastQC

Normal level of sequence duplication in 20 million read mammalian sample



Data Quality Assessment - FastQC

Normal sequence bias at beginning of reads due to non-random hybridization of random primers

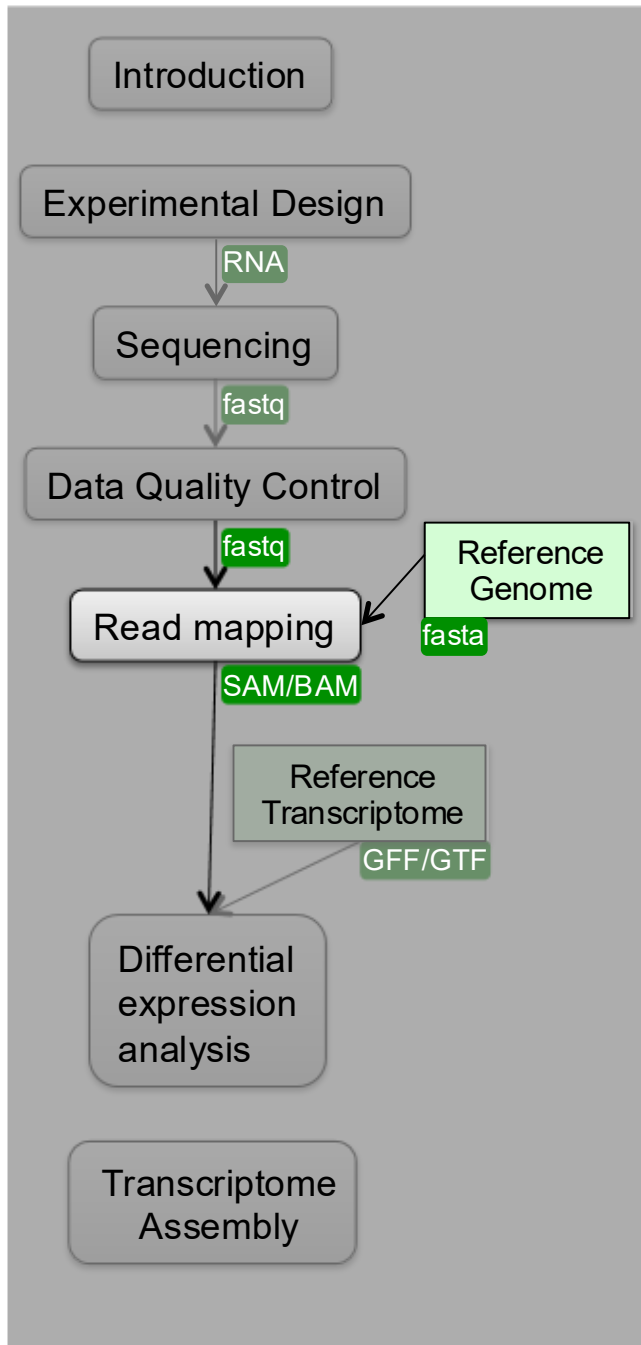


Data Quality Assessment

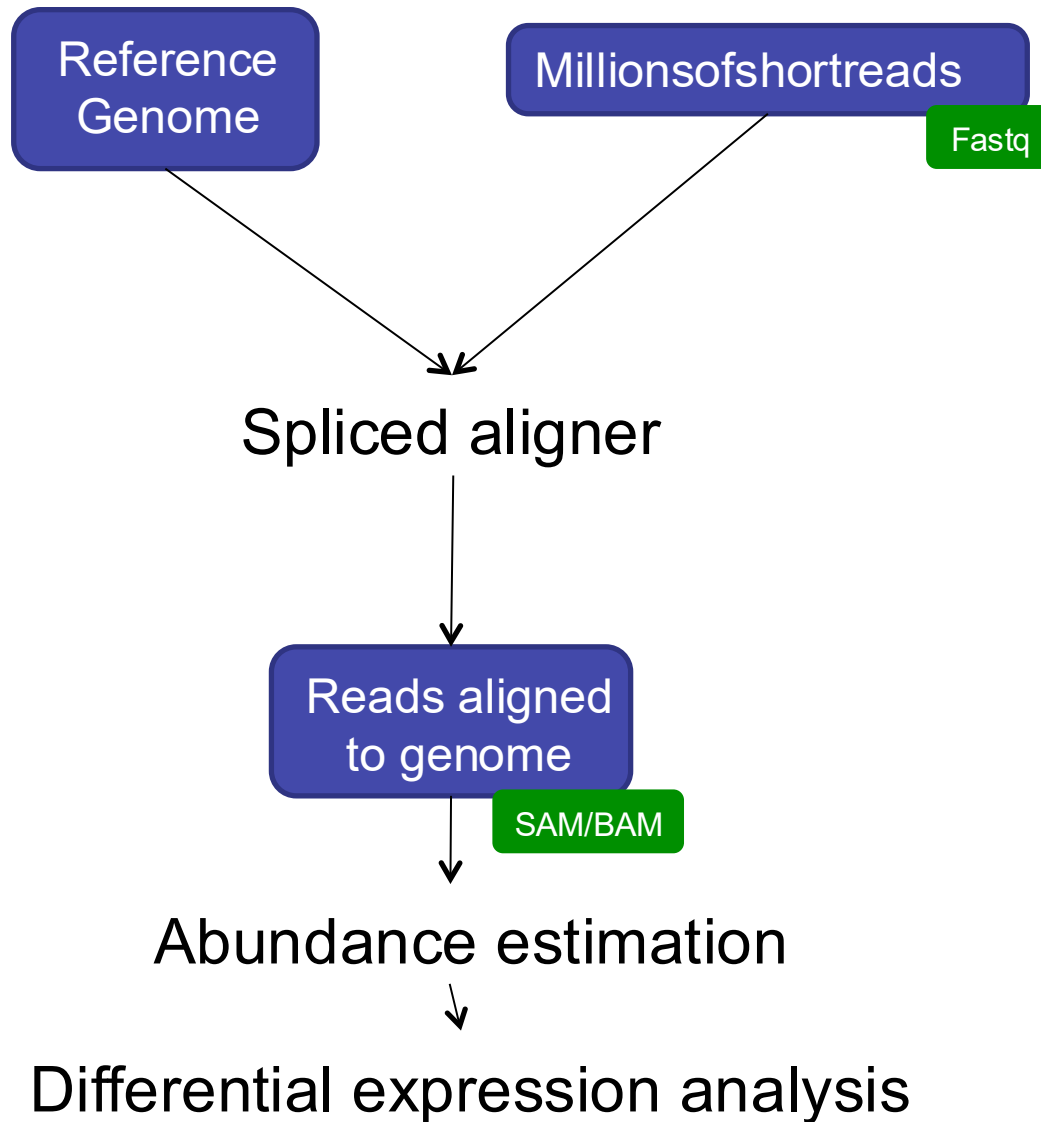
- Recommendations
 - Generate quality plots for all read libraries –
Trim and/or filter data if needed
 - Always trim and filter for de novo transcriptome assembly
 - Regenerate quality plots after trimming and filtering to determine effectiveness

Read Mapping

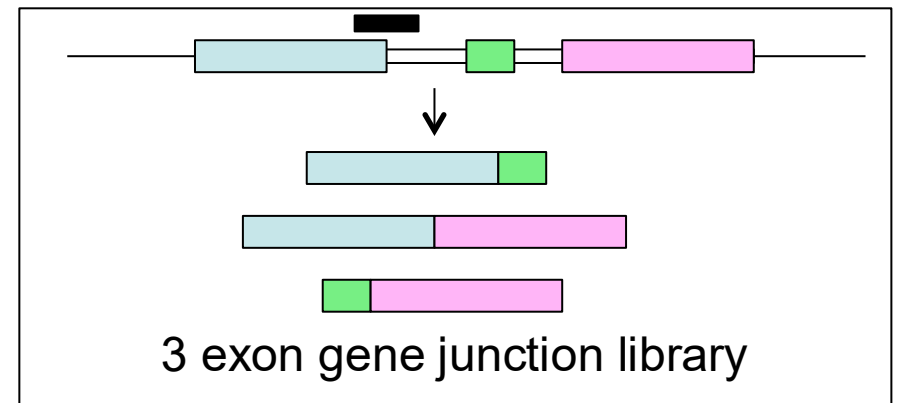
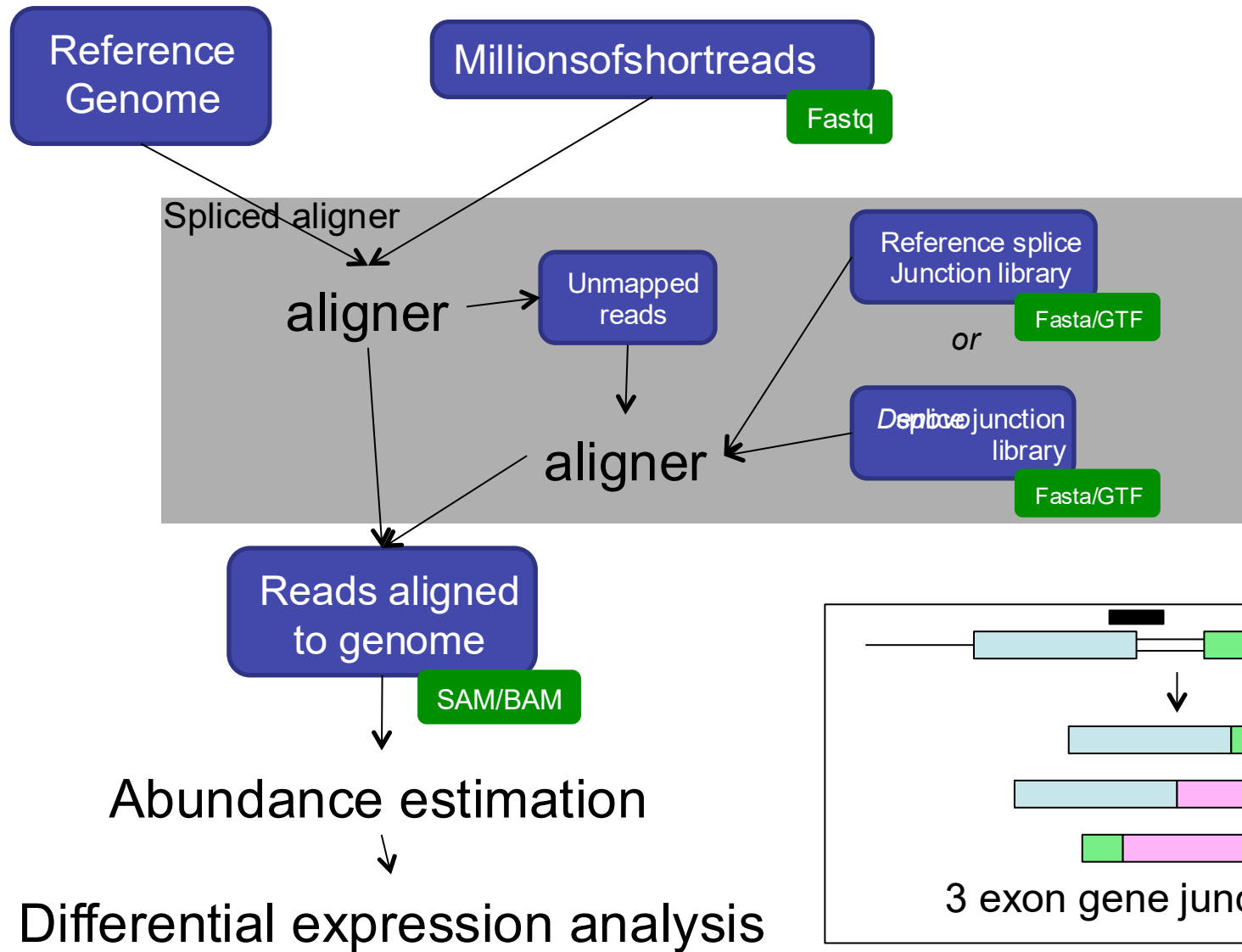
- Pipeline
- Software
- Input
- Output



Mapping – with reference genome



Mapping – with reference genome

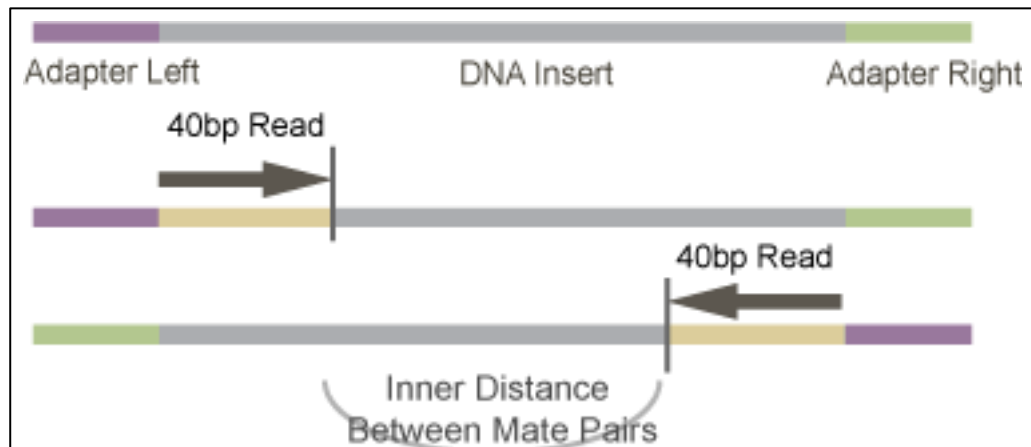


Mapping

- Alignment algorithm must be
 - Fast
 - Able to handle SNPs, indels, and sequencing errors
 - Allow for introns for reference genome alignment (spliced alignment)
- Burrows Wheeler Transform (BWT) mappers
 - Faster
 - Few mismatches allowed (< 3)
 - Limited indel detection
 - Spliced: Tophat, MapSplice
 - Unspliced: BWA, Bowtie
- Hash table mappers
 - Slower
 - More mismatches allowed
 - Indel detection
 - Spliced: GSNAP, MapSplice
 - Unspliced: SHRiMP, Stampy

Mapping

- Input
 - Fastq read libraries
 - Reference genome index (software-specific: /project/db/genomes)
 - Insert size mean and stddev (for paired-end libraries)
 - Maplibrary(orasubset) using estimated mean and stddev
 - Calculate empirical mean and stddev
 - Galaxy: NGSPicard:insertionsizemetrics
 - Cufflinks standarderror
 - Re-map library using empirical mean and stddev

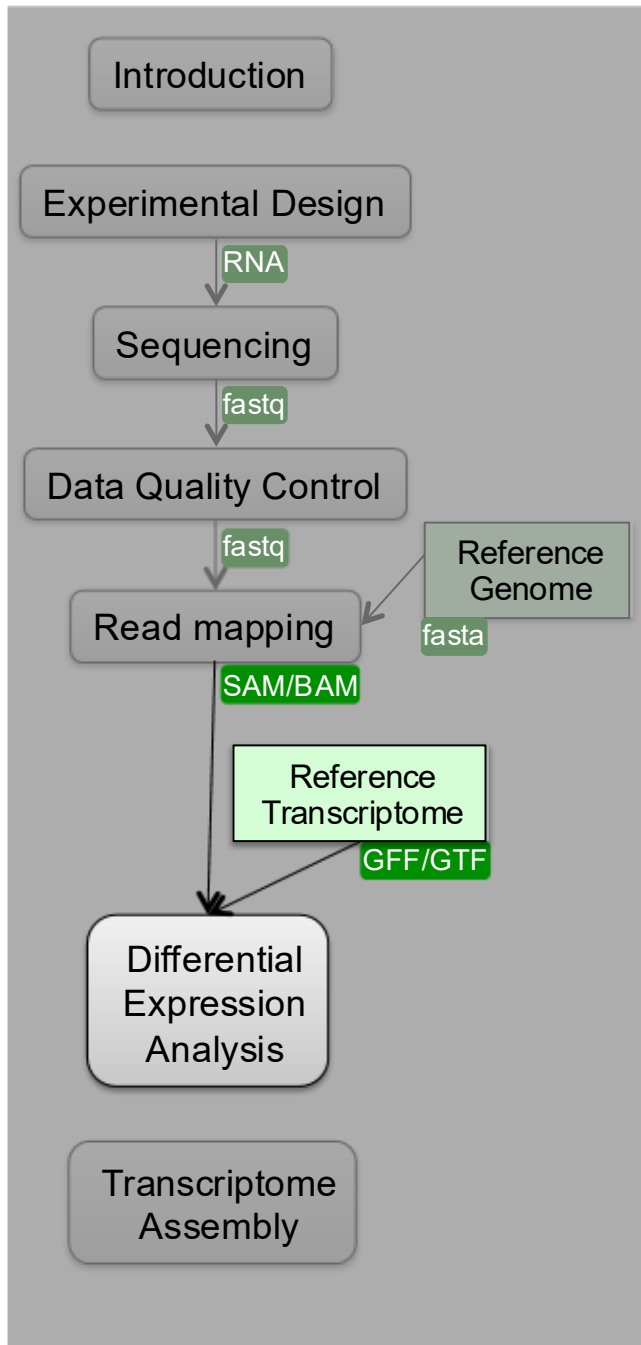


Mapping

- Output
 - SAM(text) / BAM (binary) alignment files
 - SAMtools – SAM/BAM file manipulation
 - Summary statistics (per read library)
 - % reads with unique alignment
 - % reads with multiple alignments
 - % reads with no alignment
 - % reads properly paired (for paired-end libraries)

Differential Expression

- Discrete vs continuous data
- Cuffdiff and EdgeR



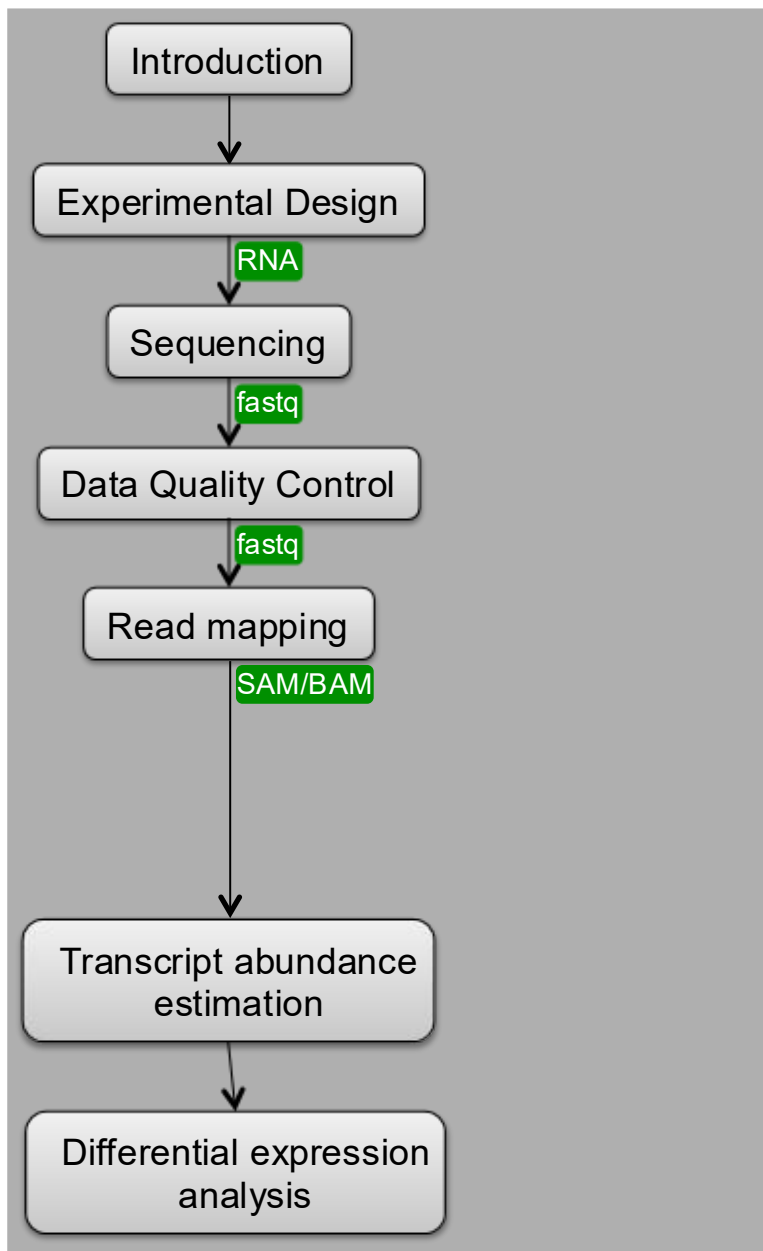
Differential Expression

- Discrete vs Continuous data
 - Microarray fluorescence intensity data: continuous
 - Modeled using normal distribution
 - RNA-Seq read count data: discrete
 - Modeled using negative binomial distribution

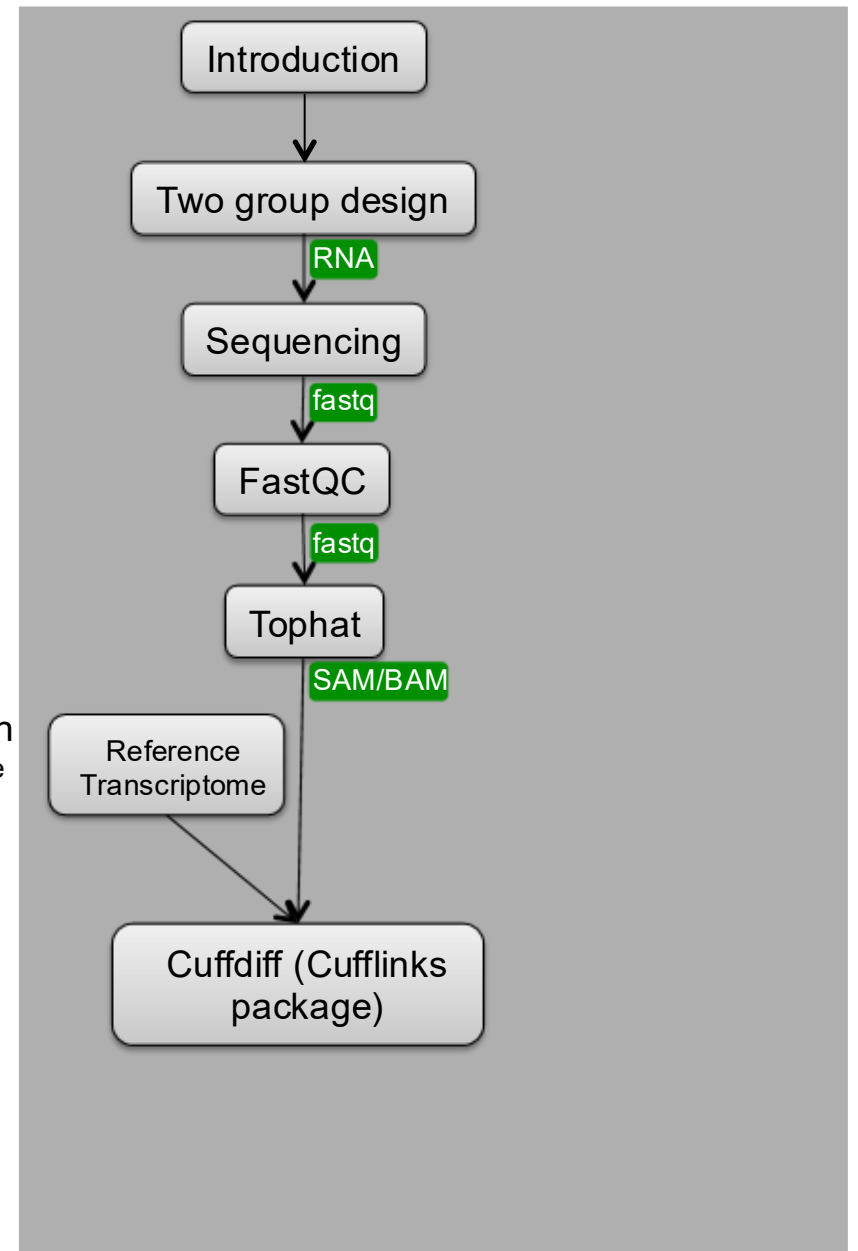
Microarray software cannot be used to analyze RNA-Seq data

Differential Expression

- Cuffdiff(Cufflinkspackage)
 - Pairwise comparisons – Differential gene, transcript, and primary transcript expression; differential splicing and promoter use
 - Easy to use, well documented
 - Input: transcriptome, SAM/BAM read alignments (abundance estimation built-in)

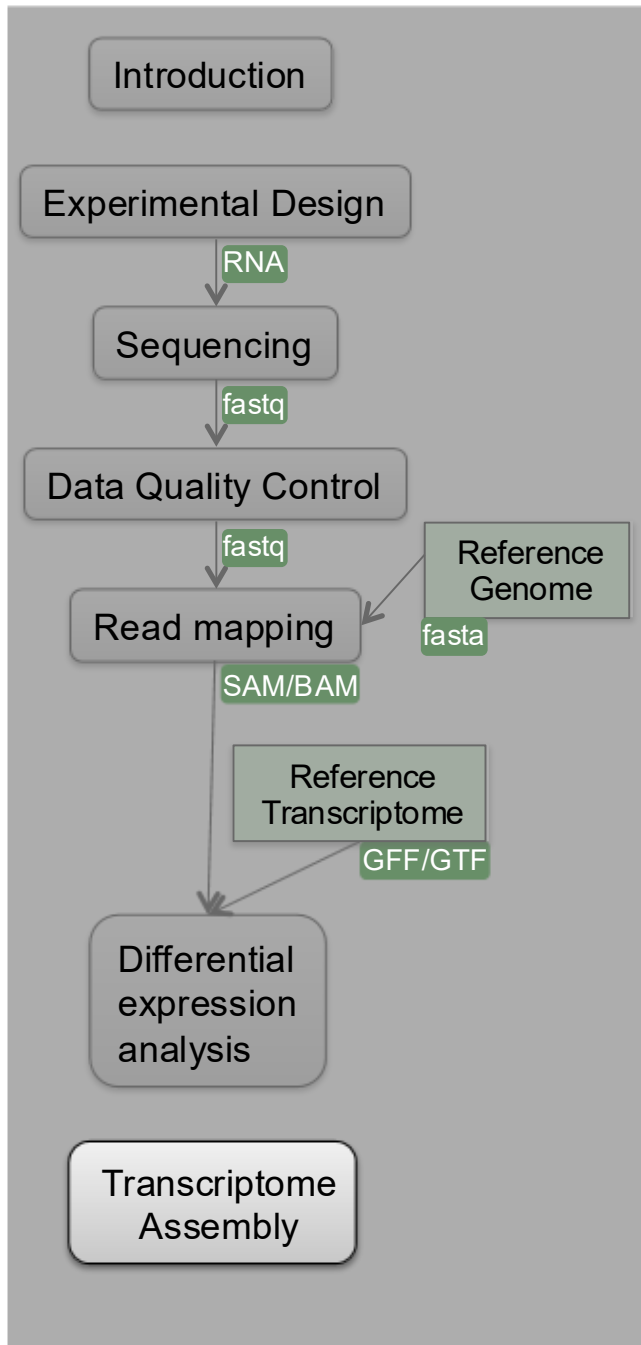


→
Suggested
Implementation
(referencegenome
available)



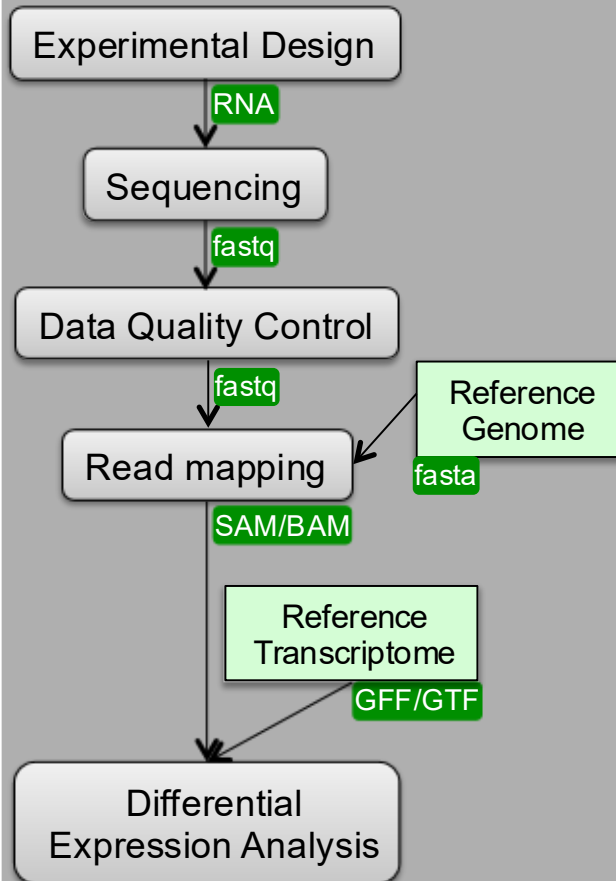
Transcriptome Assembly

- Pipeline
- Software
- Input
- Output



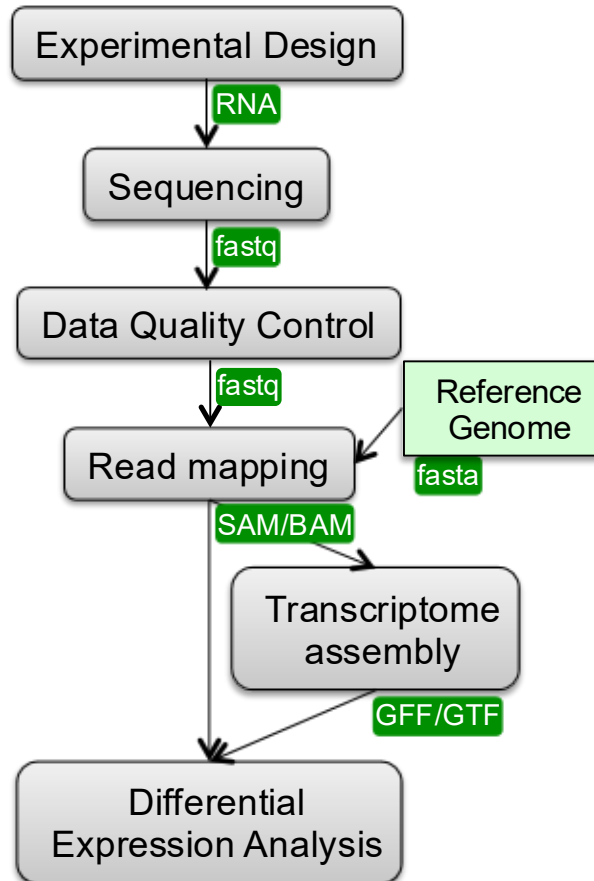
RNA-Seq

- Reference genome
- Reference transcriptome



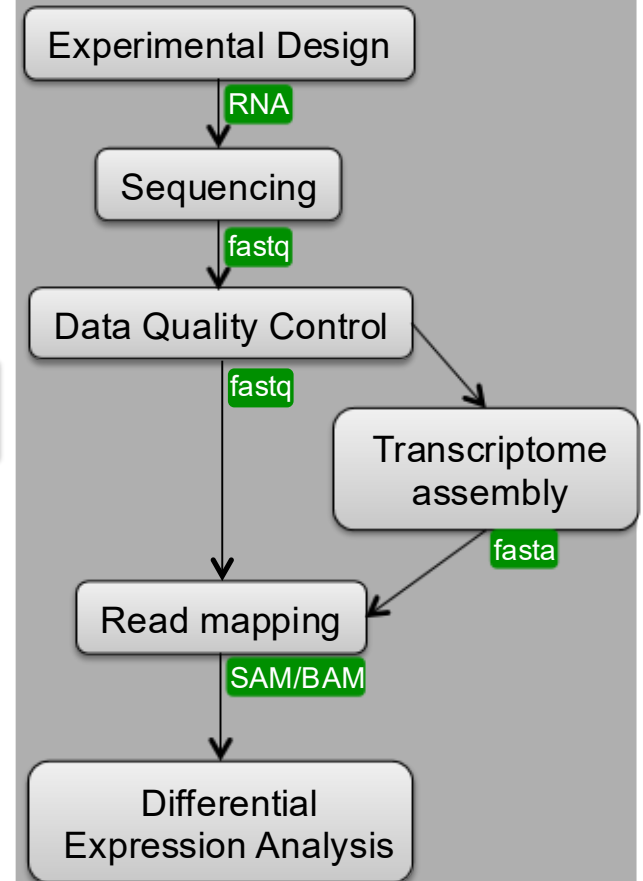
RNA-Seq

- Reference genome • No reference transcriptome



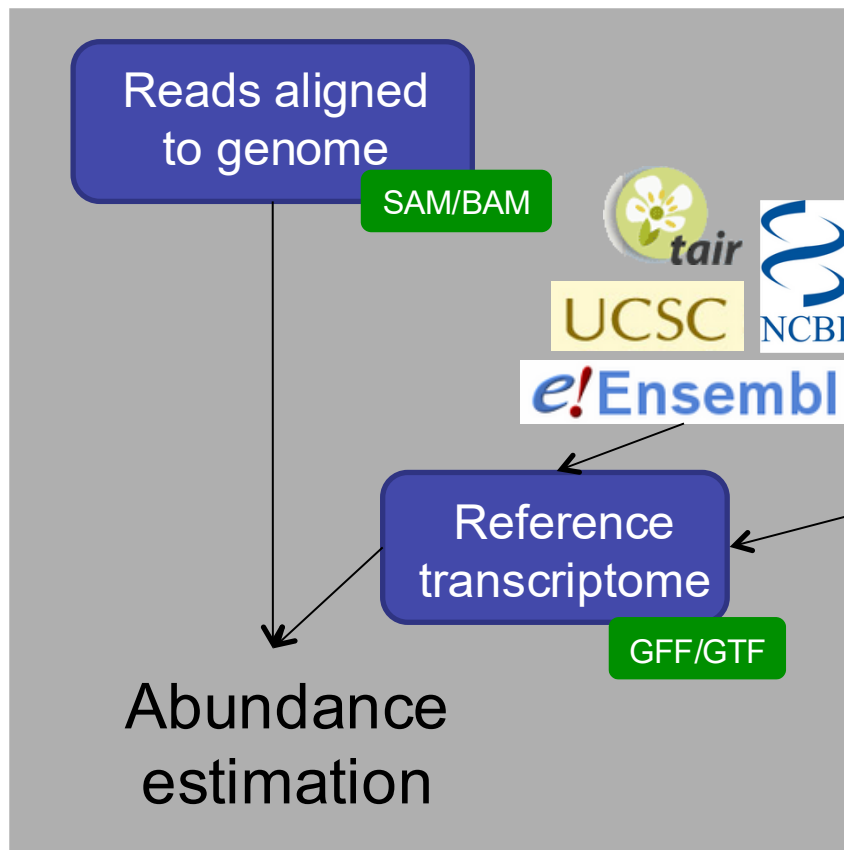
RNA-Seq

- No reference genome • No reference transcriptome

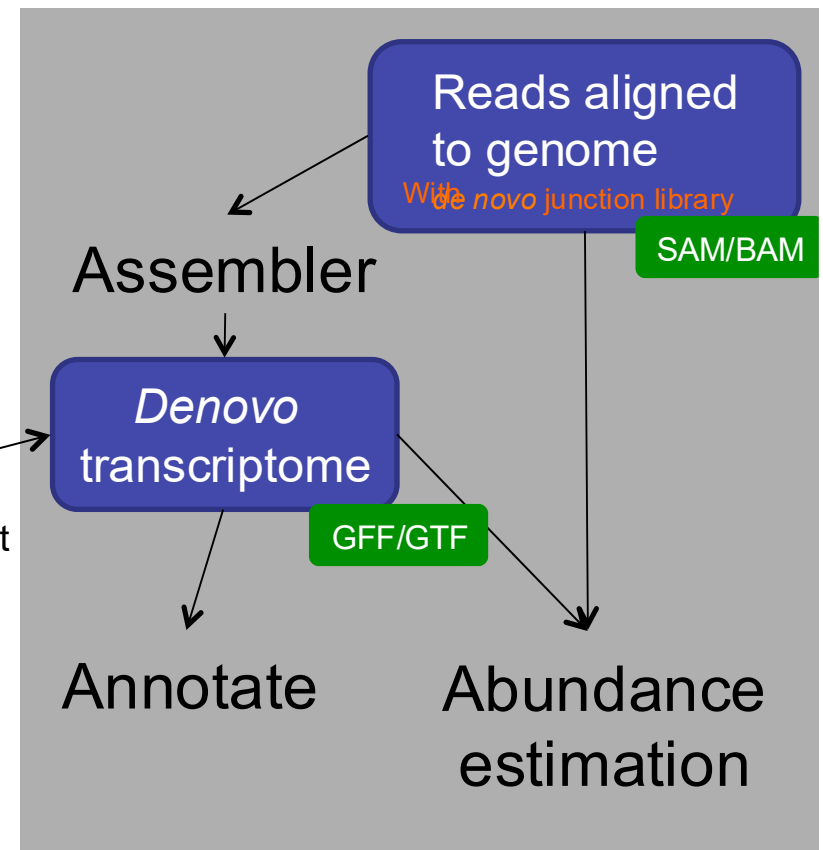


Transcriptome Assembly -with reference genome

Reference transcriptome available



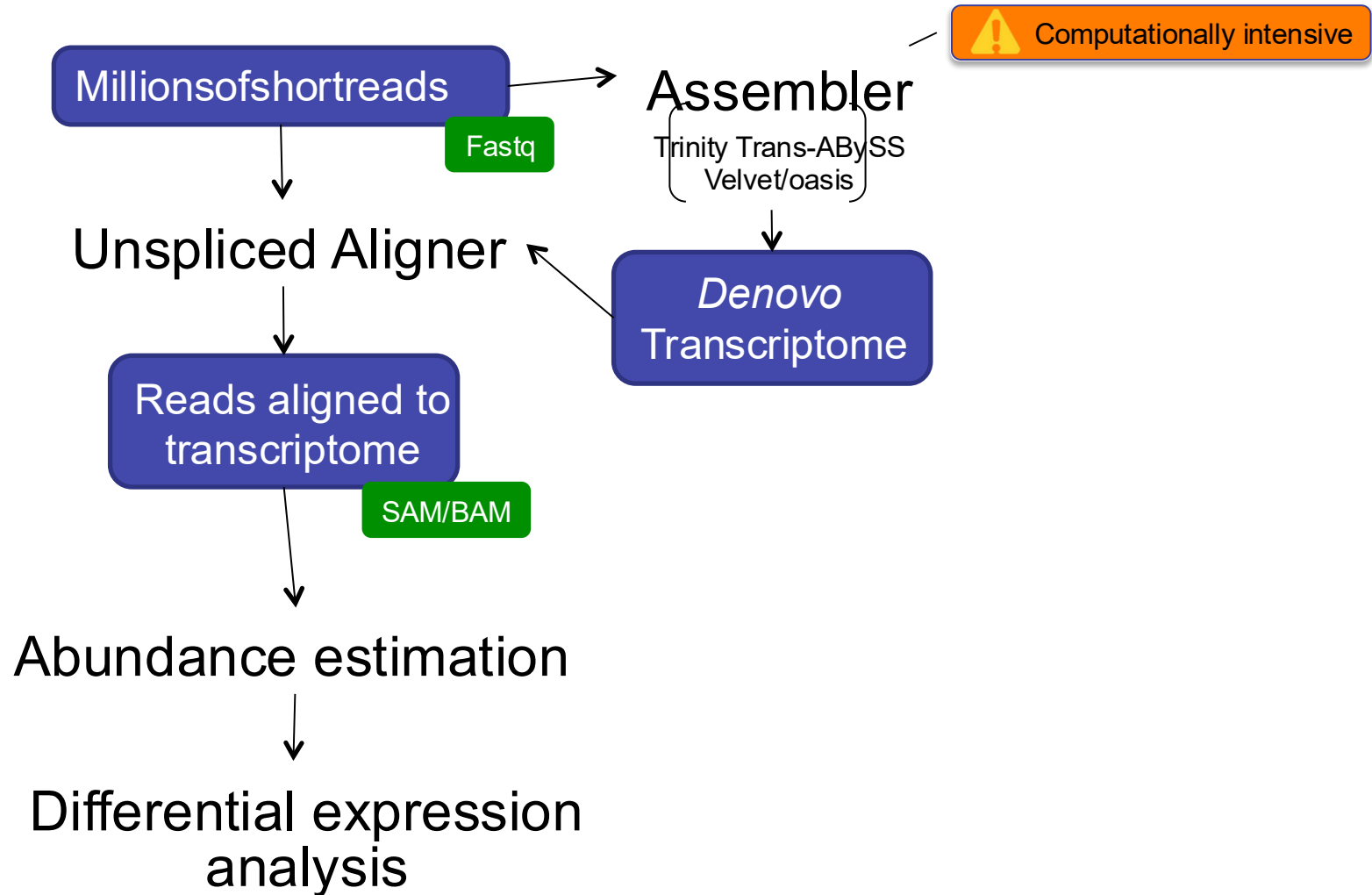
No/poor reference transcriptome available



Transcriptome Assembly -with reference genome

- Reference genomebasedassembly
 - Cufflinks, Scripture
- Reference annotation based assembly
 - Cufflinks
- Transcriptome comparison
 - Cuffcompare
- Transcriptome Annotation
 - Generate cDNA fasta from annotation (Cufflinks'gffreadprogram)
 - Align to library of known cDNA (RefSeq, GenBank)

Transcriptome Assembly –no reference genome



Further Reading

Bioinformatics for HighThroughputSequencing

Rodríguez-Ezpeleta, Naiara.; Hackenberg, Michael.; Aransay, Ana M.;
SpringerLink New York, NY : Springer c2012

Online access through ULibrary

RNA sequencing: advances, challenges and opportunities

Fatih Ozsolak¹ & Patrice M. Milos¹ Nature Reviews Genetics 12, 87-98 (February 2011)

Computational methods for transcriptome annotation and quantification using RNA-seq

Manuel Garber, Manfred G. Grabherr, Mitchell Guttman & Cole Trapnell Nature Methods 8, 469–477 (2011)

Table of RNA-Seq software

Next-generation transcriptome assembly

Jeffrey A. Martin & Zhong Wang Nature Reviews Genetics 12, 671-682 (October 2011)

Differential gene and transcript expression analysis of RNA-seq experiments with TopHat and Cufflinks

Cole Trapnell, Adam Roberts, Loyal Goff, Geo Perteira, Daehwan Kim, David Kelley, Harold Pimentel, Steven Salzberg, John L. Rinn & Lior Pachter

Nature Protocols 7, 562–578 (2012)

SEQanswers.com

Popular bioinformatics forums

biostar.stackexchange.com